Technological Forecasting & Social Change 189 (2023) 122353

Contents lists available at ScienceDirect

Technological Forecasting & Social Change

journal homepage: www.elsevier.com/locate/techfore

Identifying technology opportunity using SAO semantic mining and outlier detection method: A case of triboelectric nanogenerator technology

Xin Li [a], Yundi Wu [a], Haolun Cheng [b], Qianqian Xie [c], Tugrul Daim [d,e,*]

[a] College of Economics and Management, Beijing University of Technology, Beijing, China
[b] College of New Media, Beijing Institute of Graphic Communication, Beijing, China
[c] Centre for Science and Technology Studies (CWTS), Leiden University, Leiden, Netherlands
[d] Portland State University, Portland, USA
[e] Chaoyang University of Technology, Taiwan

ARTICLE INFO

Keywords:
Technology opportunity
Subject-action-object semantic analysis
Outlier detection
Triboelectric nanogenerator
Text mining
Technology forecasting

ABSTRACT

With the high integration of science and technology development, how to early identify technology opportunity is crucial for the governments' and enterprises' research and development (R&D) strategic planning and innovation policy to gain a first-mover advantage in the market competition environment. Most researchers have applied Subject-Action-Object (SAO) semantic mining approach or outlier detection method to mine scientific papers or patent information for identifying technology opportunity. However, few researchers have combined information from both scientific papers and patents to identify technology opportunity by integrating SAO semantic mining and outlier detection method. Therefore, this paper proposes a research framework that uses scientific papers and patents as data resources, and integrates SAO semantic mining and outlier detection method to identify technology opportunity. In this framework, we first use the SAO semantic mining method to mine technical problems and solutions contained in scientific papers and patents respectively. Then we conduct comparative analysis to identify potential technology opportunity in the gaps between scientific papers and patents. Secondly, we use a outlier detection method to identify outlier points in scientific papers, and we incorporate the outlier points into the analysis scope of technology opportunity identification. Finally, we combine the results of SAO semantic mining method with outlier detection method, and use expert knowledge to identify technology opportunity. The triboelectric nanogenerator technology is selected as a case study to verify the feasibility of this framework. The results show that the framework can effectively and comprehensively identify technology opportunity from the two levels of technical problems and technical solutions. This paper contributes to technology opportunity study, and will be of interest to triboelectric nanogenerator technology R&D experts.

1. Introduction

In recent years, the emergence and development of new technologies have not only influenced global economic and social development, but also changed people's lifestyles (Rifkin, 2011). Most businesses devote a significant percentage of their budget to support technological innovation, and the identification of technology opportunities is a starting point for managing the subsequent activities (Yoon and Magee, 2018). Technology opportunity is the intelligence information for decision-makers that can adapt to future market competition, technological innovation and social changes (Kerr et al., 2006; Cozzens et al., 2010). How to early identify technology opportunity is crucial for the governments' and enterprises' research and development (R&D) strategic planning and innovation policy to gain a first-mover advantage in a competitive market. Therefore, several scholars have conducted a large number of methods to identify technology opportunities.

The concept of technology opportunity was first proposed by Porter, who believes that technology opportunity analysis is to predict the forthcoming technology form or technology development trend by mining the horizontal and vertical development trend of existing technologies in a specific field and their mutual relationships (Porter and Detampel, 1995). The traditional methods of identifying technology opportunities such as Delphi method (Cho and Lee, 2013), morphological analysis method (Yoon, 2008; Gu and Huang, 2009; Lee et al., 2014;

* Corresponding author at: Engineering and Technology Management Department, Portland State University, Portland, USA.
E-mail addresses: lixinyz@bjut.edu.cn, lixinbjut@gmail.com (X. Li), q.xie@cwts.leidenuniv.nl (Q. Xie), ji2td@pdx.edu, tugrul.u.daim@pdx.edu (T. Daim).

https://doi.org/10.1016/j.techfore.2023.122353
Received 8 July 2022; Accepted 17 January 2023
Available online 1 February 2023
0040-1625/© 2023 Elsevier Inc. All rights reserved.

Feng et al., 2019; Lee and Lee, 2019; Zhang and Wen, 2020) are mostly qualitative analysis, mainly based on expert knowledge and experience. However, qualitative methods rely too much on expert judgement, and the results have certain deviations (Huang et al., 2016). Moreover, the explosive growth of data in the era of big data, especially the rapid growth of scientific papers and patents data that contain massive amounts of technical information, makes it difficult to identify technology opportunities only using expert judgement (Li et al., 2019). Therefore, the use of objective data and quantitative analysis methods, supplemented by expert knowledge for technology opportunity identification has attracted wide attention from scholars. Some scholars use scientific papers and patents as data resources, then apply various methods such as text mining, citation network clustering, or keywords clustering to compare and analyze topics in scientific papers and patents. Simultaneously, scholars have broadened the definition of technology opportunity based on Porter's definition. Some scholars consider that technology blank area contains the technology opportunity (Jia et al., 2017; Yoon et al., 2018). Some scholars believe that technology opportunities exist in the integration of technical knowledge, and based on this, they build technical knowledge networks and use link prediction methods to predict the connection possibilities between unconnected nodes in the network, then consider the most potential connections as technology opportunities (Kim et al., 2017; Park and Yoon, 2018; Han et al., 2019). Some scholars look for technologies that enable companies to develop new technologies or new products from patents. They use comprehensive analysis to identify technology opportunities based on the enterprises' own technological capabilities (Sun et al., 2015; Seo et al., 2016; Park and Yoon, 2017; Kim et al., 2018; Lee and Lee, 2019; Choi et al., 2019; Roh et al., 2019; Lee et al., 2020). However, these methods have some limitations in analyzing technology opportunities. Existing studies often use keywords analysis in text mining because keywords include the information and characteristics of the text, which can reveal the meaning of the text. However, keywords cannot express the semantic relationship of the text content. Therefore, the keyword analysis method cannot express the topics of the text well. As a result, the granularity of technology opportunities identified by keyword analysis method is insufficient. The identified technology opportunities are macro-level in nature with the majority being technology trends or new technologies. In addition, in the definition of technology opportunity, some scholars define technology opportunity as gaps between science and technology (Shibata et al., 2010a). Technologies that exist in the scientific papers and do not exist in the patents are identified as technology opportunities (Shibata et al., 2010a; Shibata et al., 2010b; Shibata et al., 2011; Wang et al., 2015; Ogawa and Kajikawa, 2015). Scholars have also developed a number of quantitative methods to identify technology opportunities. However, the technology opportunities identified by these methods are also more macro-level, with the majority of them being at the level of technology trends or new technologies.

The Subject-Action-Object (SAO) structure can effectively display the semantic relationship of sentence elements and express a complete semantics information. When describing the technology, the SAO structure also displays the semantic relationship of the technologies (Choi et al., 2011). Such as, in the SAO structure, S can represent the technical resolution, and AO can represent the technical problem (Moehrle et al., 2005). In this way, the SAO structure indicates the technical "problem-resolution" format. Therefore, the SAO semantic mining method overcomes the limitations of keyword analysis method, and is used by some scholars to identify technology opportunities (Huang et al., 2016). However, the analysis results in these studies cannot fully cover the possible technology opportunities because they only analyze the main technical content of the clusters obtained by using SAO structure semantic analysis method. Isolated or distinctive points outside the clusters are not considered. While some emerging or novel technologies often do not have enough time to accumulate knowledge to form representative clustering topics (Wang et al., 2015). Thus, the isolated or distinctive points might represent the most novel technologies in technical fields and lead to further technological development trends (Yoon and Kim, 2012; Shi et al., 2019a). Therefore, these methods may lose important technical information, leading to incomplete identification of technology opportunities. At the same time, previous research use SAO structure semantic analysis method to mine and identify technology opportunities only stay at the level of technical problems, and does not focus on technology opportunities at the level of technical solutions.

In practice, the most important intelligence information for R&D managers and policy makers should include technology opportunities at the level of technical problems, technical trends, or new technologies, as well as technical solutions to existing technical problems. However, few scholars conduct research on the identification of technology opportunities at the level of technical solutions. With the rapid advancement of science and technology, the process from scientific discovery to technology development and commercialization is becoming shorter. Scientific discoveries may breed new technical problems or trends, as well as solutions to existing technical problems. Focusing on future technology, based on understanding the relationship between science and technology, identifying technology opportunities from the gaps between science and technology has become a key task for R&D managers and policy makers (Shibata et al., 2010a). Therefore, in this paper, we attempt to identify technology opportunities at the level of technical problems and technical solutions in the gaps between science and technology based on the definition of technology opportunity proposed by Shibata et al. (2010a). We try to deeply analyze the technical information represented in the SAO structure and to incorporate the analysis of isolated or distinctive points into the scope of technology opportunity identification. We propose a research framework that uses scientific papers and patents as data resources, and that applies SAO semantic mining and outlier detection method to identify technology opportunity. In this framework we first separately extract the SAO structures in the text of scientific papers and patents, and calculate the similarities between the SAO structures, and then we calculate the similarities of the scientific paper texts and patent texts respectively. According to the similarities of scientific papers and patents, multi-dimensional scaling (MDS) is performed, and different clusters are divided to obtain technical topics in the scientific papers and patents. Then we analyze the semantic relations contained in the SAO structures, focusing on analyzing the relationship between technical solution and technical problem under each topic. Secondly, we analyze the differences between the topics of scientific papers and patents, and identify potential technology opportunities from the two levels of technical problems and technical solutions. Thirdly, we use outlier detection method to identify outlier points in scientific papers. We then incorporate the outlier scientific papers into the analysis scope of technology opportunities identification, which can more comprehensively identify technology opportunities. Finally, we combine the results of SAO structure semantic analysis with the results of outlier detection method, and use expert knowledge to identify technology opportunities. The triboelectric nanogenerator technology is selected as a case study to verify the feasibility of this framework.

The rest of this paper is organized as follows. Section 2 presents the literature review. Section 3 provides the proposed methodology. Section 4 analyzes the case study of triboelectric nanogenerator technology. Section 5 concludes the paper.

2. Literature review

2.1. Gaps between science and technology analysis for identifying technology opportunity

Scientific research is considered as the source of technology and innovation in the linear innovation model (Turney, 1991; Shibata et al., 2010a). In this model, scientists create the seeds of innovation,

companies take up these seeds, develop technologies, and then industrialize (Shibata et al., 2010a). Some scholars believe that this linear innovation model is overly simplistic, ignoring the feedback loop between science and technology as well as the impact of technology on science (Turney, 1991; Williams and Edge, 1996). Science and technology are interdependent (Meyer, 2000). There are, however, gaps between science and technology. Scientists are frequently unaware of what the potential applications of their scientific discoveries are, or what companies might be interested in them. Similarly, companies usually do not realize what scientific discoveries may be valuable to them. Therefore, some scholars define the gaps between science and technology as technology opportunities (Shibata et al., 2010a; Shibata et al., 2010b). Following that, some scholars attempt to find technology opportunities in the gaps between science and technology using the definition of technology opportunity. Some scholars compare the citation network clustering of scientific papers and patents, and identify topics with scientific activity, but no technological application as technology opportunities (Shibata et al., 2010a; Shibata et al., 2010b). Some scholars use text mining and cluster analysis to identify technical topics in scientific papers and patents, and identify technical topics with scientific activity but no technological application as technology opportunities (Wang et al., 2015). Some scholars use text mining methods to extract the topics from scientific papers and patents, and then combine expert knowledge to identify topics that exist in scientific papers but not in patents as technology trends (Li et al., 2019). These findings also demonstrate that there is a technology opportunity in the gaps between science and technology. At the same time, scholars also find that advanced scientific research is one of the important sources of technology opportunities (Wang et al., 2015). Therefore, in this paper, we use scientific papers and patents as data resource to identify technology opportunities in the gaps between science and technology.

However, in previous studies to identify technology opportunities in the gaps between science and technology, scholars have mainly focused on the mainstream clusters of scientific papers and patents. While isolated points or outliers outside of clusters are not included in the research scope of technology opportunities identification. To some extent, isolated points or outliers may represent novel technologies that can lead to further technological development trends (Shi et al., 2019a), which may represent significant technology opportunities. This makes it insufficient to identify technology opportunities only by analyzing the mainstream clusters of scientific papers and patents, and may result in the loss of important technology opportunities information, resulting in incomplete technology opportunities identification results. Therefore, in this paper, we will pay attention to the outliers outside the clusters of scientific papers and patents in the process of identifying technology opportunities from the gaps between science and technology, and incorporate them into the analysis scope of technology opportunity identification.

2.2. SAO semantic mining

Sleator and Temperley (1995) establish the Subject-Action-Object (SAO) structure by extracting subjects, actions and objects from English sentences based on the analysis of English grammar. The subject (S) represents the executor of the action, the action (A) represents the action of the subject, and the object (O) represents the recipient of the action. The SAO constitute the most fundamental elements of a sentence. While expressing sentence semantics, the SAO structure can also effectively display the semantic relationship among sentence elements, allowing for the expression of complete semantics (Choi et al., 2011). Some scholars combine the TRIZ with functional analysis, and use the SAO structures to represent the functional relationship between various components in a technology (Cascini et al., 2004). The SAO structure can represent a large amount of technology information, and can effectively reflect the key concepts and structures of components in a technology and their relationships. In the SAO structure, S represents a technology, and AO represents the function of the technology, so that the SAO structure represents the technology-function relationship. For example, in the sentence "Triboelectric Nanogenerator can harvest biomechanical energy.", "Triboelectric Nanogenerator" is the subject, "harvest" is the action, and "biomechanical energy" is the object. "S" denotes a specific technology, while "AO" represents the function of the technology. Furthermore, the SAO structure can be organized in a problem-resolution format if the S represents the technical resolution and the AO represents the technical problem (Moehrle et al., 2005). In the SAO structure, S and O can represent technical components. If A is a partitive verb such as "have", "be made of", and "composed of", then the component of a subject may include the object as a component (Choi et al., 2011). So the SAO structure can represent the inclusion relationships among technical components or technologies (Choi et al., 2011). It is because the SAO structure can represent a variety of technology information, and can be automatically extracted from patents and scientific papers using natural language processing method, the SAO structure semantic analysis method has been widely used in the research field of technology opportunity identification.

Some scholars use the SAO structure semantic relations to construct a semantic network or technology form matrix to identify technology opportunities (Choi et al., 2011; Wang et al., 2017; Yang et al., 2018; Han et al., 2019; Li et al., 2020b). Some scholars identify technical problems based on extracting the SAO structure in the titles and abstracts of patents, and identified potential technology opportunities by constructing a technology roadmap (Li et al., 2018). Some scholars use the similarity of SAO structure to measure the similarity of patents in order to find patents with high similarity, to draw patent cluster maps, and to further identify technology opportunities through analysis of technological evolution trends (Yoon and Kim, 2012; Wang et al., 2019).

However, when using the SAO semantic analysis method to identify technology opportunities, few scholars have conducted in-depth analysis of the technical information represented by the SAO structure, especially the technology-function relationship, problem-resolution relationship, and inclusion relationship among technical components expressed by SAO structure. In previous research, when SAO semantic analysis method was used to identify technology opportunities, most of the research focused on technical problems or technologies to achieve a certain function. There is no in-depth analysis of solutions to these technical problems or technologies to achieve a certain function, which leads to the identified technology opportunities that are not deep and detailed enough.

Therefore, in this paper, when identifying technology opportunities, we consider a large amount of technical information represented by SAO structure, as well as deeply analyze the information of technology-function relationship, problem-resolution relationship, and inclusion relationship among technical components represented by SAO structure.

2.3. Outlier detection

In statistics, an outlier is an observation that differs significantly from other observations. At present, outlier detection has been mainly applied in fraud detection, fault diagnosis, and communication detection. The outlier detection methods are also diverse. According to the scope of detection, outlier detection methods can be divided into two categories: local outlier detection and global outlier detection. The most common local outlier detection methods are K-Nearest Neighbor (KNN) algorithm and Local Outlier Factor (LOF) algorithm (Alghushairy et al., 2020). KNN outlier detection is the most common distance-based unsupervised outlier detection algorithm. The algorithm obtains the outlier of the point by calculating the distance from each data point to k nearest neighbor points. If the size of the outlier exceeds a preset threshold, the data point is judged to be abnormal (Dang et al., 2015). The density-based LOF detection algorithm measures the difference between the local density of each point and the local density of its k neighbors. If the value of LOF is greater than 1, the point is judged to be an abnormal

point (Breunig et al., 2000). These two algorithms are simple to calculate without the need for training with labels, but the selection of K value is a difficult problem. In addition, as the dimensionality of data increases, computational complexity and time complexity will increase. About the global outlier detection methods, the widely used algorithms include K-means algorithm, Support Vector Machines (SVM) and Isolation Forest (IF). K-means algorithm clusters the data, and the data far from the clustering center are determined as outlier points (Kenaza et al., 2018; Macqueen, 1967). This algorithm does not require pre-training, but its results depend too much on the effect of clustering, so it is not the most ideal outlier detection algorithm. Some scholars propose an outlier detection algorithm based on SVM, which using SVM to classify raw data to determine abnormal points (Khan et al., 2007). Some scholars propose another outlier detection algorithm based on the division idea, namely IF (Liu et al., 2012). They use a binary tree structure namely iTree to isolate abnormal data. Anomalies are more likely to be divided closer to the root of iTree, while normal points are more likely to be divided deeper in iTree. Anomalies are these points with short average path length. This method takes full advantage of the unusual "fewer but different" characteristics, and it is more effective than the commonly used basic distance and density measurements. Moreover, the algorithm has low linear time complexity and small memory requirements (Liu et al., 2012).

Some scholars believe that unusual technologies are technology opportunities, and outliers usually represent relatively new patents, enabling developers to discover future innovative research directions early (Yoon and Kim, 2012). The idea of these research is to find the most novel/unique patents in the patent set, and to use indicators or expert knowledge to evaluate the development value of the patents. Unique patents with development value are identified as technology opportunities. Some scholars use the local anomaly detection LOF algorithm to detect novel points, heterogeneous points or outliers in the patent groups, and to calculate their novelty and opportunity value to determine technology opportunities (Yoon and Kim, 2012; Lee et al., 2015; Kim et al., 2017). Some scholars use the global outlier detection algorithm to identify abnormal points in patents for identifying technology opportunities (Wang et al., 2015; Huang et al., 2016). The local outlier detection algorithm pays more attention to the points close to the target point and less to the global distribution. Global outlier detection is the opposite. Few scholars have used both methods together. In this study, in order to avoid the shortcomings of only using local outlier detection algorithms or global outlier detection algorithms, two widely commonly used local outlier detection algorithms KNN and LOF, and the global outlier detection algorithm with the best performance Isolation Forest (IF), are separately selected according to the data types to detect outlies together.

2.4. Comparative analysis of related work

Based on the above literature review, we can conclude that previous studies on technology opportunity identification have the following shortcomings. First, most scholars only analyze the main contents of the clusters of scientific papers and patents when mining the gaps between science and technology. While isolated points or outliers outside of clusters are not included in the research scope of technology opportunities identification, this leads to incomplete technology opportunities identification results. Second, in the process of technology opportunities analysis, scholars mostly use the SAO semantic analysis method for text clustering analysis instead of the rich semantic information contained in the SAO structure itself to conduct in-depth technology opportunities mining. Since the text clustering analysis yields relatively broad topics, the identified technology opportunities are mostly at the level of technical problems. As a result, the technology opportunities identified by this method have a higher granularity, and the technology opportunities at the technical solution level are not thoroughly identified. Third, when using the outlier detection algorithm to identify outliers, scholars only use the local outlier detection method or the global outlier detection method, which may lead to incomplete outlier identification results, and this in turn leads to the incomplete identification results of technology opportunities.

In order to make up for the above shortcomings of the research on the identification of technology opportunities, in this paper, we will try to seek technology opportunities in the gaps between science and technology, and identify technology opportunities at the two levels of technical problems and technical solutions. First, we analyze the information of technology-function relationship, problem-resolution relationship, and inclusion relationship among technical components represented by SAO structure, and then we deeply analyze the technical problem and technical solution information in the gaps between science and technology. Second, we combine the local outlier detection and the global outlier detection method to detect outlier points that are outside the clusters. Then we will analyze the evolution of outlier points, and incorporate them into the analysis scope of technology opportunity identification. Finally, we propose a research framework that employs SAO semantic mining and outlier detection method to identify technology opportunities using scientific papers and patents as data resources.

3. Methodology

The relationships between science and technology have long been debated in academia (Li et al., 2019). Most scholars believe that science and technology co-evolve and interact in complex ways (Turney, 1991; Williams and Edge, 1996; Meyer, 2000), and that science and technology work as dancing partners (Rip, 1992). Some empirical evidences, however, show that there are still gaps between science and technology. Some scholars contend that there is a technology opportunity in the gaps between science and technology, and they define the technical topics with scientific activity but no technological application as technology opportunities (Shibata et al., 2010a). Many scholars have used case studies to demonstrate that there are indeed technology opportunities in the gaps between science and technology (Shibata et al., 2010b; Wang et al., 2015; Li et al., 2019). Therefore, in this paper, we attempt to use scientific papers and patents as data resources to identify technology opportunity in the gaps between science and technology. We try to expand the definition of technology opportunity proposed by Shibata et al. (2010a), and to conduct in-depth analysis of the gaps between science and technology. The main idea of identifying technology opportunity is as shown in Fig. 1.

On the one hand, in Fig. 1(a), scientific papers and patents may contain technical problems. These technical problems may promote the emergence of new technologies or trends in technological development. In area A, technical problems are developed in both scientific papers and patents. In area B, some technical problems appear in scientific papers, but not in patents. The situation in area C is opposite to that in area B. Area D is where neither scientific papers nor patents focus on technical problems. Technological problems in area B may be novel technical knowledge, which could give birth to new technologies or new trends in technological development. It could be a technology opportunity at the level of a technical problem. On the other hand, scientific papers and patents may contain information on technical solutions to existing technical problems. In area F, technical solutions appear both in scientific papers and in patents. In area E, some technical solutions have only appeared in scientific papers and have not yet appeared in patents. The situation in area H is opposite to that in area E. The area G is where neither scientific papers nor patents focus on technical solutions. Technical solutions in area E may be new solutions to existing technical problems, which could be technology opportunities at the technical solution level. Therefore, in this paper, we attempt to identify technology opportunities at the level of technical problems in area B and those at the level of technical solutions in area E

In order to deeply mine the technology opportunity information that

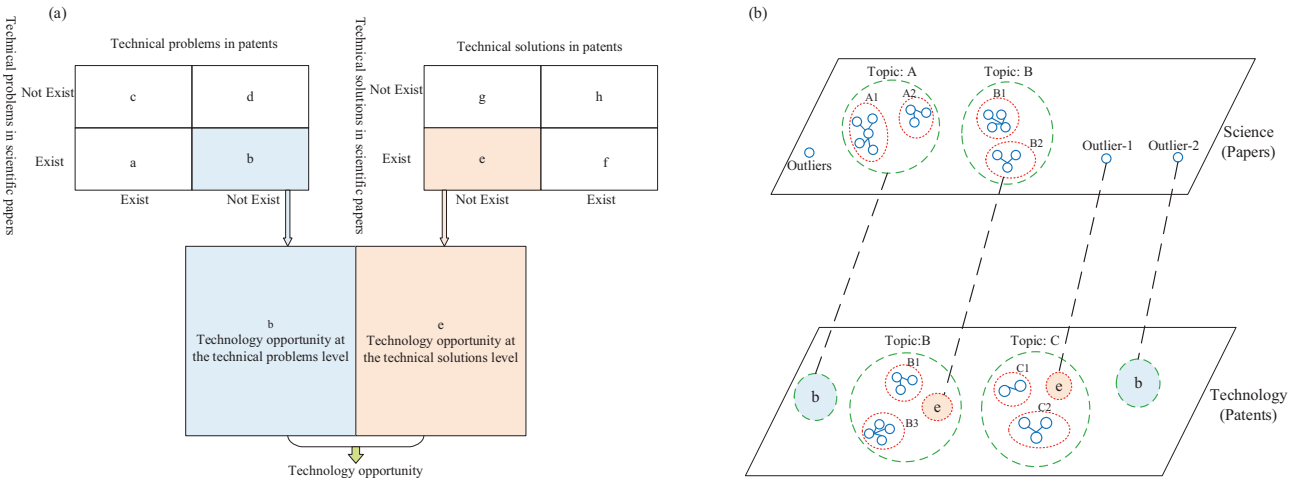


Fig. 1. Relationships between science and technology. (a) Two types of technology opportunity; (b) Gaps between science and technology. Source: Modified from Shibata et al. (2010a).

may be contained in scientific papers and patents, we use SAO semantic mining method to analyze technical topics in scientific papers and patents, and the outlier detection method to identify outliers in scientific papers. Since the SAO structures can represent a variety of technology information, we refer to the topics of scientific papers and patent clustered by the semantic similarity of the SAO structures as first-level technical topics. These first-level technical topics are generally macro technical topics, that expressing generalized technical information. We name them as technical problems. After obtaining the first-level technical topic, we conduct a detailed analysis of the technical information contained in the first-level technical topic according to the technology information represented by the SAO structures, such as technology-function, problem-resolution, and inclusion relationship among technical components information. And we refer to this different technology information as second-level technical topics. These second-level technical topics represent technical component solutions, technical functional solutions, and resolutions to technical problems. Therefore, we label them as technical solutions, i.e., technical solutions to existing technical problem. In this case, we can obtain different first-level and second-level technical topics in scientific papers and patents. Each first-level technical topic may include several second-level technical topics. For the same first-level technical topic, different second-level technical topics represent different technical solutions to the same technical problem. Outlier scientific papers may contain novel technical problems or solutions. Through the analysis of the difference between the first-level technical topics of scientific papers and of patents, as well as the difference between the topics of outlier scientific papers and the first-level technical topics of patents, we can obtain technology opportunities at the level of technical problems. Through the analysis of the difference between the second-level technical topics of scientific papers and of patents, as well as the difference between the topics of outlier scientific papers and the second-level technical topics of patents, we can obtain technology opportunities at the level of technical solutions. The specific process of identifying technology opportunity is shown in Fig. 1 (b).

In Fig. 1(b), green dashed circles represent first-level technical topics, red dashed circles represent second-level technical topics, and small blue circles denote a scientific paper or a patent. Scientific papers outside of the first-level technical topics are outliers. The first-level technical Topic A only exists in scientific papers and does not exist in patents. It could be a technology opportunity at the level of technical problem, which corresponds to area B in Fig. 1(a). The first-level technical Topic B in scientific papers contains two second-level technical Topics B1 and B2, and it in patents contains two second-level technical Topics B1 and B3. B1 exists in both scientific papers and patents, corresponding to area f in Fig. 1(a). B3 exists only in patents and not in scientific papers, corresponding to the h region in Fig. 1(a). B2 only exists in scientific papers and does not exist in patents, corresponding to area e in Fig. 1(a). B2 could be a technology opportunity at the level of technical solution. The outlier scientific paper Outlier-1 contains technical solutions, which belongs to patents' first-level technical Topic C, but no corresponding patents' second-level topic has been found. Therefore, the topic of Outlier-1 corresponds to area e in Fig. 1(a), which could be a technology opportunity at the level of technical solution. The outlier scientific paper Outlier-2 contains technical problems, but no corresponding patents' first-level topic has been found. Therefore, the topic of Outlier-2 corresponds to area b in Fig. 1(a), which could be a technology opportunity at the level of technical problem.

Based on the above main ideas of identifying technology opportunity, we propose a research framework that use scientific papers and patents as data resources and apply SAO semantic mining and outlier detection method to identify technology opportunity. The whole process of the framework is shown in Fig. 2, which can be divided into four stages: First, we collect data of scientific papers and patents related to triboelectric nanogenerator technology, and preprocess the data. Second, we use the SAO semantic mining method to extract the SAO structures from the scientific papers and patents year by year, and calculate the similarities between scientific papers and scientific papers, as well as the similarities between patents and patents. We determine the first-level technical topics and second-level technical topics in scientific papers and patents through the MDS and SAO structure interpretation, and then we construct a technology tree. Third, since some points outside the scientific papers' clusters, we use KNN algorithm, LOF algorithm, and IF algorithm to detect outlier points in the scientific papers year by year, and take the intersection of the results of the three algorithms as the final robust outlier scientific papers. Fourth, we conduct a comparative analysis of the topics of scientific papers and patent each year to find out the first-level topics that exist in scientific papers but do not exist in patents, and those second-level topics that exist in scientific papers for two years or more but do not exist in patents. Then, we compare topics of the outlier scientific papers with the first-level and second-level topics of patents to find the topics that exist in outlier scientific papers but do not exist in patents. Finally, we combine the above results with expert knowledge to comprehensively identify technology opportunities.

The detailed analysis steps of this framework are as follows:

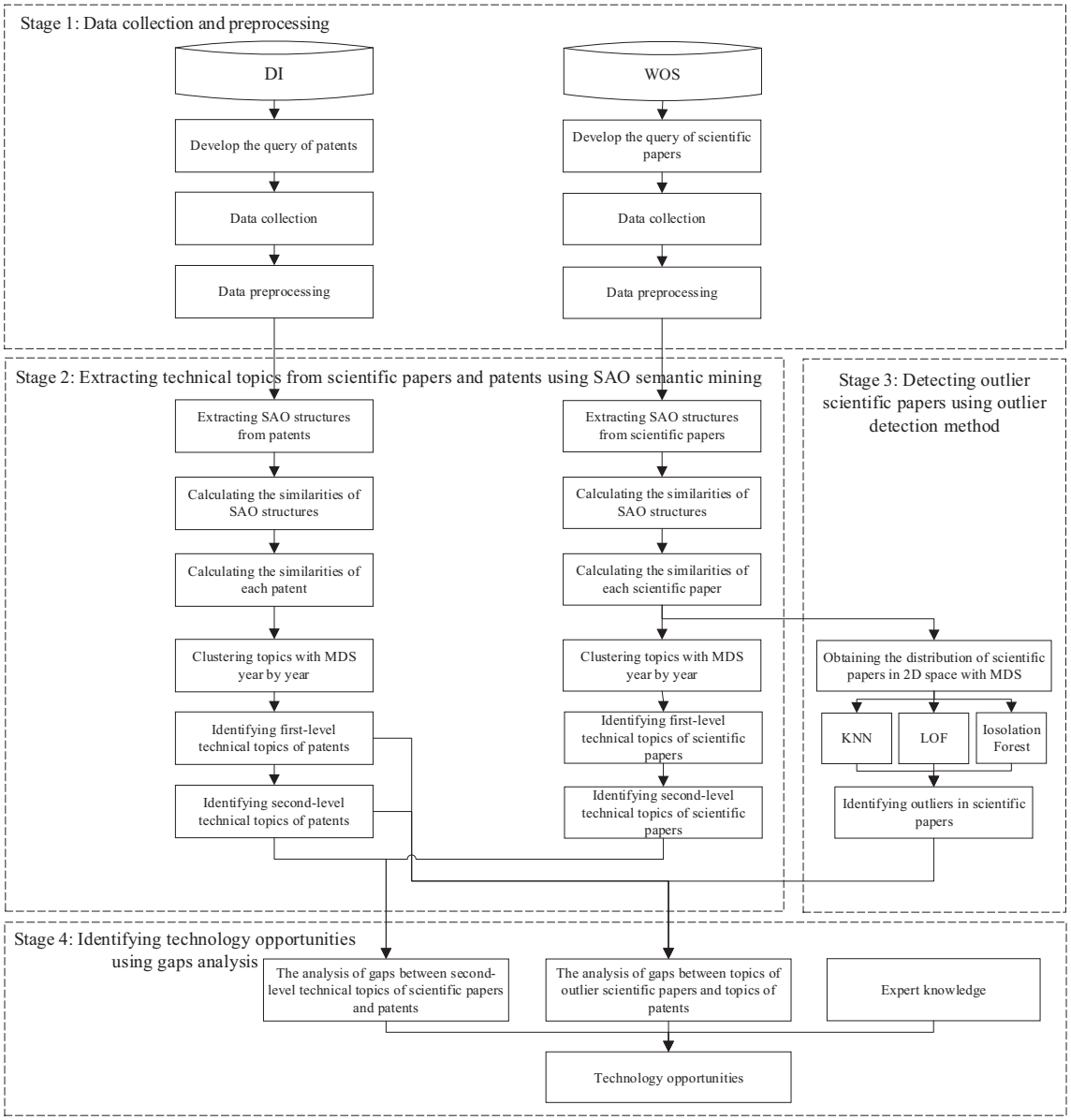


Fig. 2. The framework for identifying technology opportunity using SAO semantic mining and outlier detection method.

3.1. Stage 1: data collection and preprocessing

We use Web of Science (WOS) and Derwent Innovation (DI) databases as data resources to collect data, and use search queries related to the research topics to download relevant scientific papers and patents. Then we clean the noise data unrelated to the triboelectric nanogenerator and transform the preprocessed data into a format suitable for text mining, and finally divide the preprocessed data by year.

3.2. Stage 2: extracting technical topics from scientific papers and patents using SAO semantic mining

Step 1: Extracting the SAO structures from scientific papers and patents

StanfordCoreNLP is an open source natural language processing toolkit. We use it to extract SAO structures from the titles and abstracts of the scientific papers and patents by Python.

Step 2: Calculating the similarity of SAO structures

Semantic disambiguation is a basic activity in natural language processing, which determines the quality of subsequent analysis. We use the semantic disambiguation algorithm in the WordNet toolkit to find the most appropriate word meaning of the SAO structures, and to perform semantic disambiguation on the extracted SAO structures. Then we use the similarity algorithm to calculate the semantic similarities of the disambiguated SAO structures. The SAO semantic similarity matrices can be obtained.

Step 3: Calculating the similarity of texts

Based on the similarities of each SAO structures, we calculate the similarities of scientific papers and the similarities of patents respectively. The calculation formula is as follows:

$$sim(p_i, p_j) = \frac{2*NUM_{SAO(p_i,p_j)}}{NUM_{SAO(p_i)} + NUM_{SAO(p_j)}} \quad (1)$$

When the similarity value between each SAO structure exceeds a predefined threshold, the two SAO structures are considered to be

similar. In formula (1), $NUM_{SAO(p_i,p_j)}$ represents the number of similar SAO structures in the paper (or patent) p_i and paper (or patent) p_j, $NUM_{SAO(p_i)}$represents the number of SAO structures in the paper (or patent) p_i and $NUM_{SAO(p_j)}$represents the number of SAO structures in the paper (or patent) p_j.

Step 4: Extracting the first-level technical topics of scientific papers and patents.

In order to extract the technical topics in scientific papers and in patents, we use the MDS analysis method to reduce the dimensionality of the similarity relationships of scientific papers and of patents each year and map them into a two-dimensional space. In the two-dimensional space, each scientific paper or patent is mapped to a point, and we obtain the coordinate information of each scientific paper or patent in two-dimensional space. Scientific papers or patents with similar semantics are relatively close in two-dimensional space, and they form a cluster in two-dimensional space. Therefore, by observing the distribution of points in a two-dimensional map, we group scientific papers or patents with similar semantics into a cluster. Finally, based on the clustering results of the scientific papers and patents, combined with the interpretation of the SAO structures, we obtain the first-level technical topics of each cluster and name them as the technical problems. We then obtain the first-level technical topics of the scientific papers and the first-level technical topics of the patents.

Step 5: Constructing a technology tree in the field of triboelectric nanogenerator based on experts' knowledge.

In order to obtain a whole picture of triboelectric nanogenerator technology and the relationships between technical problems and technical solutions, we need to build a technology tree of triboelectric nanogenerator technology, which will provide a basis for identifying technology opportunities from the aspects of technical problems and technical solutions. Therefore, two domain experts, comprising one senior researcher who has focused on triboelectric nanogenerator studies for more than three years from the College of Materials Science and Engineering, Beijing University of Technology, China, and one research assistant are invited to serve as our panel to construct the technology tree of triboelectric nanogenerator. We treat the first-level technical topics in the scientific papers and patents as objective evidence for our panel to construct the technology tree of triboelectric nanogenerator.

Step 6: Extracting the second-level technical topics of scientific papers and patents.

After we obtain the whole picture of triboelectric nanogenerator technology, we conduct an in-depth analysis of the content of each scientific paper cluster and patent cluster obtained in the fourth step. According to the first-level technical topic, we use the information expressed by the SAO structures, such as technology-function, problem-resolution, and inclusion relationship among technical components information, to further classify the technical information contained in the first-level technical topics in depth and detail. Then, we extract different technical information for each technical problem, which we called second-level technical topics, i.e., different solutions to technical problems. Finally, we obtain second-level technical topics in scientific papers and patents.

3.3. Stage 3: detecting outlier scientific papers using outlier detection method

In the Stage 2, we obtain the coordinate information of each year's scientific papers in two-dimensional space. In this stage, in order to detect outlier scientific papers in each year's scientific papers, we input the coordinate information into the outlier detection algorithm to identify outliers in each year's scientific papers. Since the advantages and disadvantages of each outlier detection algorithm are different, the use of multiple outlier detection algorithms for outlier detection can reduce the deviation caused by using only one outlier detection algorithm (Yoon and Kim, 2012). Therefore, we use two local outlier detection algorithms KNN and LOF, and the global outlier detection algorithm IF to detect outliers together. In this paper, we make the intersection of the results of these three algorithms, and only consider outlier scientific papers that are identified as outliers by the three algorithms.

The details of these three algorithms are as follows.

3.3.1. KNN algorithm

The KNN outlier detection method is a distance-based outlier detection algorithm. The basic idea of the algorithm is to calculate the KNN distance between each data in the data set and other data, and sort the average KNN distance of each point in descending order. The first N points with the largest distance are considered as outliers (Dang et al., 2015). The KNN outlier detection algorithm has a high level of computational granularity. However, when the density of the dataset varies greatly, the KNN algorithm may miss some outliers. For a point $p(x,y)$, the k-nearest neighbor distance $D^k(p)$ between the k-th nearest neighbor point $p_k(x_k,y_k)$ is calculated as follows:

$$D^k(p) = \sqrt{(y-y_k)^2 + (x-x_k)^2} \tag{2}$$

3.3.2. LOF algorithm

The LOF is a density-based outlier detection algorithm. The core idea is that the density of outlier points should be smaller than the density of other points in the neighborhood. The *LOF* value of point p represents the degree of abnormality of the point. Generally speaking, the value of *LOF* is greater than 1, indicating that the density of this point is less than the density of its neighboring points, and the more likely this point is an outlier point. This algorithm can handle clusters of any size, and performs well on datasets with significant density differences. Like the KNN algorithm, the LOF algorithm is more sensitive to local data. The specific calculation formulas of *LOF* are as follows (Breunig et al., 2000):

$d(p,q)$ represents the distance between points p and q. For any positive integer k, the distance k of point q is expressed as $k-distance(p)$, which is defined as the distance $d(p,o)$ between p and object $o \in D$.

Given the k distance of p, the k distance neighborhood of p contains every point q whose distance from p is no more than k distance is called the k nearest neighbor of p:

$$N_{k-distance(p)}(P) = \{q \in D \backslash \{p\} \,|d(p,q) \le k-distance(p)\} \tag{3}$$

If the point p is far away from o, then the reachable distance between the two is their actual distance. However, if they are close enough, the actual distance is replaced by the k distance of o. The formula for reachable distance is as follows:

$$reach-disk_k(p,o) = max\{k-distance(o), d(p,o)\} \tag{4}$$

The parameter *MinPts* specifies the minimum number of points, $lrd_{MinPts}(p)$and $lrd_{MinPts}(o)$ represent the local reachable density of point p and point o. The local reachable density calculation formula is as follows:

$$lrd_{MinPts}(p) = 1 \Big/ \left(\frac{\sum_{o \in N_{MinPts}(P)} Reach-dist_{MinPts}(p,o)}{|N_{MinPts}(P)|} \right) \tag{5}$$

The *LOF* value of point p is the average of the ratio of the local accessibility density of p to the accessibility density of the smallest neighbor of p. The calculation formula is as follows:

$$LOF_{MinPts}(\mathrm{P}) = \frac{\sum_{o \in N_{MinPts}(P)} \frac{lrd_{MinPts}(o)}{lrd_{MinPts}(p)}}{|N_{MinPts}(P)|} \tag{6}$$

3.3.3. Isolation Forest algorithm

Isolated Forest (IF) is a tree-based division algorithm. Its core idea is that abnormal points are more likely to be divided closer to the root of iTree, while normal points are more likely to be divided deeper in iTree. The Isolation Forest algorithm, which has linear time complexity and high efficiency, focuses on the global distribution of the data set.

The Isolation Forest algorithm includes two steps: in the training phase, iTree is established by recursively separating the training set until all samples are isolated or the tree reaches a specified height. In the evaluation phase, the anomaly score of each test sample is obtained from the expected path length that will be obtained through each tree in the isolated forest. Outliers refer to the case where the average path length on iTrees is shorter. The closer the anomaly score is to 1, the more likely it is that this point is an outlier point (Liu et al., 2012).

3.4. Stage 4: Identifying technology opportunities using the analysis of gaps between science and technology

Step 1: Based on the analysis of the gaps between the topics of scientific papers and patents in the Stage 2, we identify technology opportunities in combination with the technology tree. We consider first-level topics that exist in scientific papers but not in patents as technology opportunities at the technical problem level, and consider second-level technical topics that exist in scientific papers for two years or more, but do not exist in patents as technology opportunities at the technical solution level.

Step 2: According to the outliers identified by the three outlier detection algorithms in the Stage 3, we analyze the gaps between the topics of the identified outlier scientific papers and the topics of patents. We consider outlier scientific paper topics that do not exist in first-level patent topics as technology opportunities at the level of technical problems, and considered outlier scientific paper topics that do not exist in second-level patent topics as technology opportunities at the level of technical solution.

Step 3: We combine the technology opportunities obtained in the above two steps with the growth rate of first-level technical topics, and finally identify the technology opportunities based on expert opinions.

4. Case study

The triboelectric nanogenerator (TENG) was proposed by Wang Zhonglin's team in 2012. Triboelectric nanogenerators can not only generate electricity by harvesting energy from vibrations generated by human movement, but they can also harvest large amounts of blue energy from ocean waves and convert wave energy into electricity. The invention of the triboelectric nanogenerator was awarded the Eni Prize in 2019 and the Einstein World Science Prize in 2020. The rapid development of triboelectric nanogenerator technology is believed to have a significant impact on people's lives, and has the potential to transform the existing energy industry (Li et al., 2020a, 2020b). Therefore, identifying technology opportunities in the field of triboelectric nanogenerator technology is of great significance to the strategic R&D planning of governments and enterprises.

4.1. Data collection and preprocessing

In this paper, WOS and DI databases are used as data resource. We invited a domain expert to determine search expressions related to triboelectric nanogenerator technology. We searched for scientific papers data in WOS using the term "TS=(“ triboelectric nanogenerat*” OR “TENG” OR “TENGs” OR “Triboelectric Nanogenerat*”) ", and retrieved 2204 scientific papers from 2012 to 2019. We used the term "ABD=(triboelectric ADJ nanogenerat* OR TENG OR TENGs OR Triboelectric ADJ Nanogenerat*) AND DPRY>=(2012) AND DPRY<=(2019)" to search for patents data in DI database, and a total of 174 patents were retrieved from 2012 to 2019.

In the process of preprocessing scientific papers and patents data, we deleted irrelevant papers and patents, and obtained 1,783 scientific papers and 92 patents. Then we processed the texts to remove abnormal symbols, converted it into a format suitable for text mining, and divided the preprocessed data by year. Fig. 3 shows the annual number of scientific papers and patents related to triboelectric nanogenerator technology.

4.2. Analysis of technical topics in scientific papers and patents related to triboelectric nanogenerator using SAO semantic mining

4.2.1. Analysis of the first-level technical topics in scientific papers and patents related to triboelectric nanogenerator

We used the StanfordCoreNLP to extract the SAO structures from the titles and abstracts of the preprocessed scientific papers and patents by Python respectively. Then SAO similarity calculation and text similarity calculation of scientific papers and patents were performed. We used the MDS in SPSS software to map the text similarities of scientific papers and patents into the two-dimensional space. According to the coordinate information of each scientific paper and patent in the two-dimensional space, we divided the scientific papers and patents each year into different clusters. Then, combined with the semantics analysis of the SAO structures in the scientific papers and the patents, each type of technical topic can be obtained, that is, the first-level technical topic, and each first-level technical topic represents a technical problem related to TENG. In order to obtain the technical topic of each cluster and name it, one domain expert, comprising one senior researcher who has focused on TENG studies for more than three years from the College of Materials Science and Engineering, Beijing University of Technology, China, and one research assistant were invited to serve as our panel. The clustering process of scientific papers and clustering topics in each year are shown in Figs. 1 to 8 in the Appendix I. The clustering process of patents and clustering topics in each year are shown in Figs. 9 to 16 in the Appendix I. Each point in the figures represents a scientific paper or a patent. Their numbers are displayed next to the points. Finally, we extracted nine first-level technical topics from scientific papers. They were "Theoretical research of TENG", "Types of energy harvested by TENG", "Self-powered applications of TENG", "Integration of TENG with other batteries for mixed energy Battery", "Improving the output efficiency of TENG", "Wearable TENG", "Different materials of TENG", "Hybrid NG" and "Energy storage of TENG". At the same time, we extracted eight first-level technical topics from the patents. They were "Types of energy harvested by TENG", "Self-powered applications of TENG", "Improving the output efficiency of TENG", "Wearable TENG", "Different materials of TENG", "Hybrid NG", "Energy storage of TENG" and "Structures of TENG". The seven first-level technical topics "Types of

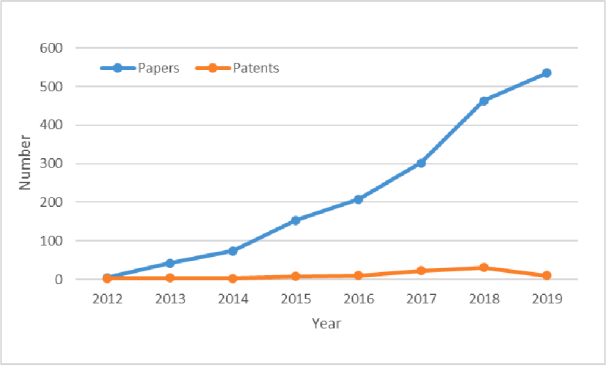


Fig. 3. Annual number of scientific papers and patents related to triboelectric nanogenerator technology.

energy harvested by TENG", "Self-powered applications of TENG", "Improving the output efficiency of TENG", "Wearable TENG", "Different materials of TENG", "Hybrid NG", and "Energy storage of TENG" were included in both scientific papers and patents. The two first-level technical topics "Theoretical research of TENG" and "Integration of TENG with other batteries for mixed energy Battery" only appeared in scientific papers, and the first-level technical topic "Structures of TENG" describing the detailed structures of TENG only appeared in patents. Therefore, after de-duplication and merging of the first-level technical topics, we finally extracted a total of ten first-level technical topics from the scientific papers and patents, and labeled the name of the first-level technical topic as shown in Table 1.

Fig. 4 shows the changes pattern of the technical topics of scientific papers in the field of TENG from 2012 to 2019, and Fig. 5 shows the changes pattern of the technical topics of patents in the field of TENG from 2012 to 2019. In Fig. 4 and Fig. 5, the circles of the same color represent the same technical topic. The size of the circle represents the number of scientific papers or patents contained in the technical topic.

It can be seen from Fig. 4 that the scientific papers' bubble chart arranged in chronological order clearly show the evolution of the technical topics in the scientific papers. The scientific papers related to TENG in the WOS database was first published in 2012. The authors demonstrated a reasonably designed bow-shaped triboelectric nanogenerator using the contact charging between a polymer film and a thin metal foil, and studied the working mechanism of TENG through finite element simulation (Wang et al., 2012). Therefore, the first-level technical topic related to TENG in 2012 was A "Theoretical Research of TENG". From 2013 to 2019, the first-level technical Topic A continued to appear in scientific papers, and more and more papers on this topic over time. Therefore, the number of scientific papers contained in the topic A gradually increased over time. The topic A is a unique topic in scientific papers. In 2013, topic B "Types of energy harvested by triboelectric nanogenerators", topic C "Self-powered applications of TENG" and topic D " Integration of TENG with other batteries for mixed energy Battery" appeared. The topic C was a research hotspot of TENG. From 2013 to 2019, the number of scientific papers contained in the topic C has increased year by year. Interestingly, in 2012, the topic C appeared in the patent, which was a year earlier than it first appeared in the scientific paper. This is because the topic C is about the application research of TENG, and the patent is more focused on the application research, so this is why the topic C appeared in the patent earlier. The topic D appeared in the scientific papers in 2013 and 2015 respectively. But after 2015, the topic D disappeared from the first-level topic of scientific papers, and we also did not find the topic D in the patent. In 2014, a new topic E "Improving the output efficiency of TENG" appeared. In 2015, two new technical topics appeared, they were Topic F "Wearable TENG" and topic G "Different materials of TENG". In 2016, the technical topics H "Hybrid NG" and I "Energy storage of TENG" appeared for the first time. There are no new technical topics between 2017 and 2019, indicating that the research direction of triboelectric nanogenerators has gradually stabilized.

Table 1
First-level technical topics in scientific papers and patents.

Topics' name	Topics	Data resources
Topic A	Theoretical research of TENG	Scientific papers
Topic B	Types of energy harvested by TENG	Scientific papers and patents
Topic C	Self-powered applications of TENG	Scientific papers and patents
Topic D	Integration of TENG with other batteries for mixed energy Battery	Scientific papers
Topic E	Improving the output efficiency of TENG	Scientific papers and patents
Topic F	Wearable TENG	Scientific papers and patents
Topic G	Different materials of TENG	Scientific papers and patents
Topic H	Hybrid NG	Scientific papers and patents
Topic I	Energy storage of TENG	Scientific papers and patents
Topic J	Structures of TENG	Patents

It can be seen from Fig. 5 that the patents' bubble chart arranged in chronological order clearly show the evolution of the technical topics in the patents. The patents related to TENG in the DI database was first published in 2012. The technical topic C "Self-powered applications of TENG" and the technical topic J "Structures of TENG" appeared in 2012. The technical topic J describes the structures and components of TENG in detail, but it has not been found in scientific papers. Although this topic contains a small number of patents, the topic J appeared six times in patents between 2012 and 2019. In 2014, the technical topics were B "Type of energy collected by triboelectric nanogenerators" and E "Improving the output efficiency of TENG". In 2015, the technical topics were B "Types of energy harvested by triboelectric nanogenerators" and C" Self-powered applications of triboelectric nanogenerators, E "Improving the output efficiency of TENG" and J "Structures of TENG". The topic F "wearable TENG", H "hybrid NG" and G "different materials of triboelectric nanogenerators" appeared for the first time in 2016. Among them, the number of scientific papers and patents included in the topic F gradually increased after 2016, and it has become an important research direction of TENG. Topic I "Energy storage of TENG" appeared for the first time in 2017. It should be noted that the topic A "Theoretical Research of TENG " in the scientific paper has not been found in patents. This also reflects the difference between scientific papers and patents, that is, scientific papers pay more attention to basic theoretical research, while patents pay more attention to applied research about TENG.

After we obtained the first-level topics in the scientific papers and patents related to TENG, we compared the topics year by year, and the comparison results are shown in Table 2. Table 2 depicts the evolution of first-level technical topics in scientific papers and patents from 2012 to 2019. This demonstrates that the description and research on TENG in scientific papers and patents are different. From Table 2, we can find that the technical topic B " Types of energy harvested by TENG " first appeared in scientific papers in 2013, and in patents in 2014. The technical topic C "Self-powered applications of TENG" first appeared in patents in 2012, and in scientific papers in 2013. That is because the patents focus on applied research. Technical topic D "Integration of TENG with other batteries for mixed energy Battery" appeared in scientific papers in 2013 and 2015. After 2015, the technical topic D was no longer mentioned in scientific papers, and it did not appear in patents. The technical topic E "Improving the output efficiency of TENG" was mentioned in both scientific papers and patents in 2014. The technical topic F "Wearable TENG" and G "Different materials of TENG" first appeared in scientific papers in 2015, and then appeared in patents in 2016. The technical topic H "Hybrid NG" appeared in both scientific papers and patents in 2016. Technical Topic I " Energy storage of TENG" was first published in scientific papers in 2016 and then appeared in patents in 2017. We can see from a comparison of the first-level technical topics in scientific papers and patents that technical topics B, F, G and I appeared first in scientific papers and later in patents, with a one- or two-years' time lag. This demonstrated that in a technology-intensive field, technical topics that exist in scientific papers but do not exist in patents can be considered as technology opportunities. Therefore, scientific papers related to TENG could breed more technology opportunities.

4.2.2. Analysis of the second-level technical topics in scientific papers and patents related to the triboelectric nanogenerator

It can be seen from Table 1 and Table 2 that some first-level topics of the clustered scientific papers and patents are the same, i.e., the technical problems are the same. However, for each technical problem, there are various technical solutions. Technology tree provides a picture of the

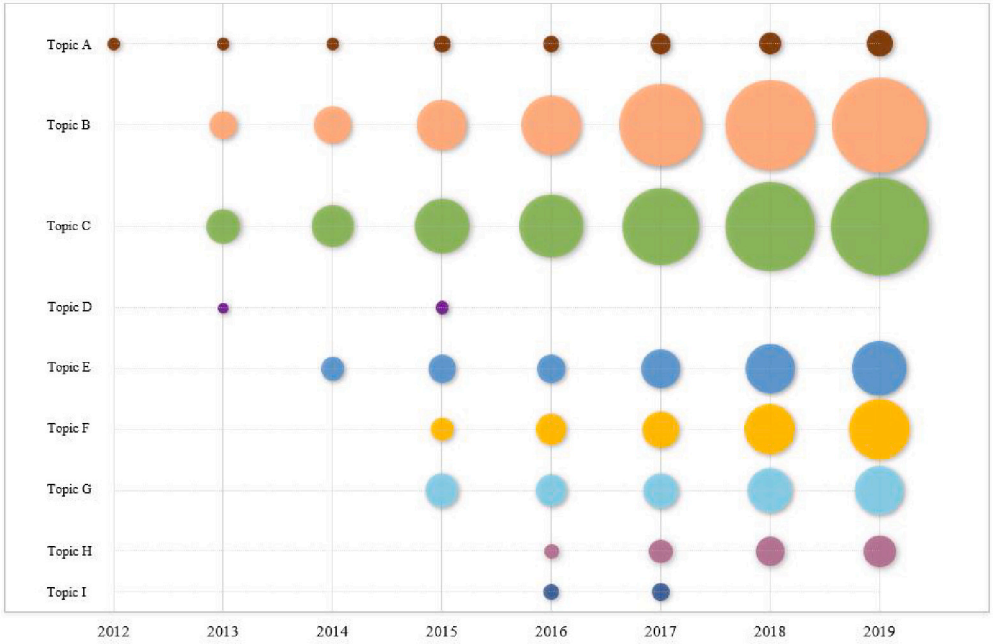


Fig. 4. The changes pattern of the technical topics of scientific papers in the field of TENG from 2012 to 2019.

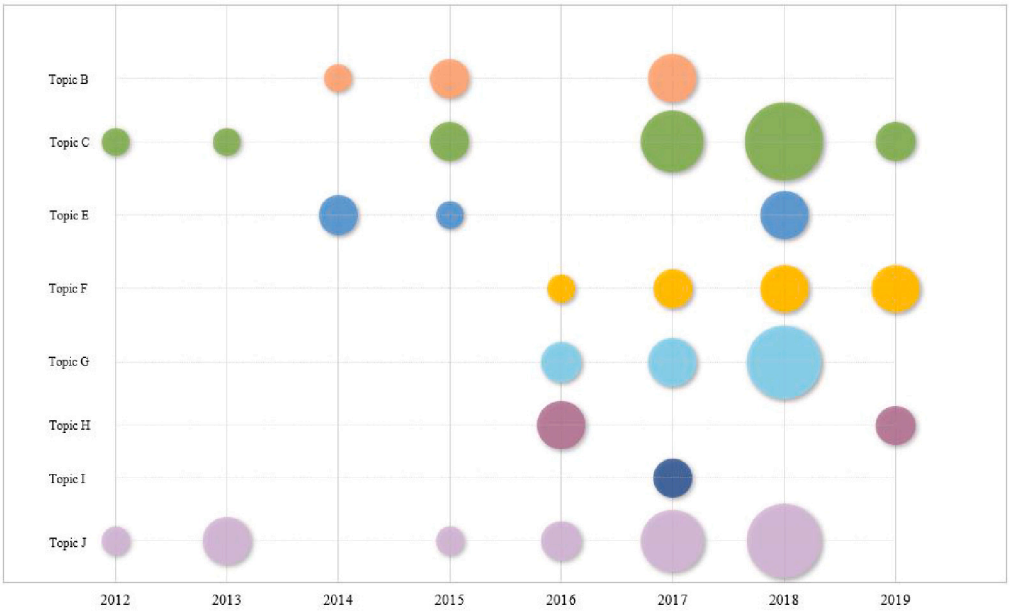


Fig. 5. The changes pattern of the technical topics of patents in the field of TENG from 2012 to 2019.

technology (Bildosola et al., 2017) to represent the relationships among technical components and technical functions of a technology in a specific technology area (Choi et al., 2012; Li et al., 2019). It is important to construct the technology tree to more fully understand the technology (Li et al., 2019). Thus, the technology tree can be used to deeply analyze the relationships between technical problems and technical solutions. In this paper, the purpose of constructing a technology tree of TENG was to provide the grounds for the classification of the first-level technical topic clustering results, and to gain a more systematic and comprehensive understanding of the whole picture of TENG. At the same time, it provides a basis for the different technical solutions identified for each first-level technical topic (i.e. technical problem), so as to provide a basis for identifying technology opportunities at the technical solution level.

As mentioned in the Methodology section, we treated the first-level technical topics in the scientific papers and patents as objective evidence for decision-making, and combined quantitative and qualitative methods for topic analysis to construct the technology tree of TENG. Hence, we engaged experts on TENG for topic analysis and classification, and we constructed the technology tree of TENG based on the first-level topics cluster of the scientific papers and patents obtained from Section 4.2.1 with the help of domain experts. The two domain experts, comprising one senior researcher who has focused on TENG studies for more than three years from the College of Materials Science and Engineering, Beijing University of Technology, China, and one research assistant were invited to serve as our panel to construct the technology tree of TENG. Fig. 6 shows the technology tree of TENG. In Fig. 6, the short name of the topic related to TENG are shown in Table 1 in the Appendix II.

Table 2
Comparison of the first-level technical topics in scientific papers and patents.

Topics	2012	2013	2014	2015	2016	2017	2018	2019
The first-level technical topics in scientific papers	A: Theoretical research of TENG	A: Theoretical research of TENG B: Types of energy harvested by TENG C: Self-powered applications of TENG D: Integration of TENG with other batteries for mixed energy Battery	A: Theoretical research of TENG B: Types of energy harvested by TENG C: Self-powered applications of TENG E: Improving the output efficiency of TENG	A: Theoretical research of TENG B: Types of energy harvested by TENG C: Self-powered applications of TENG D: Integration of TENG with other batteries for mixed energy Battery E: Improving the output efficiency of TENG F: Wearable TENG G: Different materials of TENG	A: Theoretical research of TENG B: Types of energy harvested by TENG C: Self-powered applications of TENG E: Improving the output efficiency of TENG F: Wearable TENG G: Different materials of TENG H: Hybrid NG I: Energy storage of TENG	A: Theoretical research of TENG B: Types of energy harvested by TENG C: Self-powered applications of TENG E: Improving the output efficiency of TENG F: Wearable TENG G: Different materials of TENG H: Hybrid NG I: Energy storage of TENG	A: Theoretical research of TENG B: Types of energy harvested by TENG C: Self-powered applications of TENG E: Improving the output efficiency of TENG F: Wearable TENG G: Different materials of TENG H: Hybrid NG	A: Theoretical research of TENG B: Types of energy harvested by TENG C: Self-powered applications of TENG E: Improving the output efficiency of TENG F: Wearable TENG G: Different materials of TENG H: Hybrid NG
The first-level technical topics in patents	C: Self-powered applications of TENG J: Structures of TENG	C: Self-powered applications of TENG J: Structures of TENG	B: Types of energy harvested by TENG E: Improving the output efficiency of TENG	B: Types of energy harvested by TENG C: Self-powered applications of TENG E: Improving the output efficiency of TENG J: Structures of TENG	F: Wearable TENG G: Different materials of TENG H: Hybrid NG J: Structures of TENG	B: Types of energy harvested by TENG C: Self-powered applications of TENG F: Wearable TENG G: Different materials of TENG I: Energy storage of TENG J: Structures of TENG	C: Self-powered applications of TENG E: Improving the output efficiency of TENG F: Wearable TENG G: Different materials of TENG J: Structures of TENG	C: Self-powered applications of TENG F: Wearable TENG H: Hybrid NG

The process of constructing the technology tree is as following. First, according to previous research on the classification of TENG (Li et al., 2020), we divided the TENG technology into five categories, namely energy source, application, material, structure and hybrid nanogenerator, as shown in the blue box in Fig. 6. The two domain experts and the senior researcher divided the ten first-level technical topics into the five categories. Then, we conducted an in-depth analysis of the SAO structure within each first-level technical topic, and identified the second-level technical topic of each first-level technical topic, i.e., the technical solution for each technical problem. The two domain experts and one senior researcher previewed the SAO structure within each first-level technical topic, and based on their research experience and knowledge, they consolidated interrelated the topics of SAO structure and named them by means of the most representative elements, and subdivided them to the corresponding categories and layers.

In this way, we obtained the technical solutions for each first-level technical topic, as shown in Fig. 6. In Fig. 6, the yellow box represents a type of technical solution, and the white box represents a specific technical solution. For example, under the application category of the TENG, there are technical solutions such as tracking sensor, wearable devices, electrochemical system, metal corrosion prevention, self-charging power, and other. Specific technical solutions for tracking sensor include logistics monitoring, motion tracking, and speedometers. Finally, the hierarchical structure (technology tree) of the TENG was generated, as shown in Fig. 6.

In the process of analyzing the SAO structures, we found that the relationship types represented by the extracted SAO structure can be divided into three types: technology-function relationship, problem-resolution relationship, and inclusion relationship among technical components. In the first-level technical topics, we found that the relationship type represented by the SAO structures contained in Topic C “Self-powered applications of TENG” and topic F “Wearable TENG” are mostly technology-function relationship type. For instance, the SAO structure "TENG provided self-powered motion system" contained in the first-level technical topic C "Self-powered applications of TENG", the action "provided" connects the subject "TENG" and the object "self-powered motion system" together. This SAO structure illustrates that one of the functions of the TENG is to provide electrical energy for the motion system. In the first-level technical topic C "Self-powered applications of TENG", there are some SAO structures that also represent the same function information. By analyzing these SAO structures representing the same or similar functions information together, we can summarize a function of the TENG, that is, self-powered tracking system. We referred to this function as a second-level technical topic. That is, in Table 2 in the Appendix II, the second-level technical topic "Self-powered tracking system" in the first-level technical topic C "Self-powered applications of TENG". In this paper, this type of second-level topics was referred to as technical function solutions, which address the functional aspects of TENG technology. That is to say, the second-level technical topics are the solutions of the first-level technical topic.

In the first-level technical topics, we found that the relationship type represented by the SAO structures contained in topic E “Improving the output efficiency of TENG” are mostly problem-resolution type. For example, in the SAO structure "few-layer graphene improved power", the action "improved" connects the subject "few-layer graphene" and the object "power" together. This SAO structure illustrates that few-layer graphene can improve the output efficiency of TENG. In the first-level technical topic E "Improving the output efficiency of TENG", there are

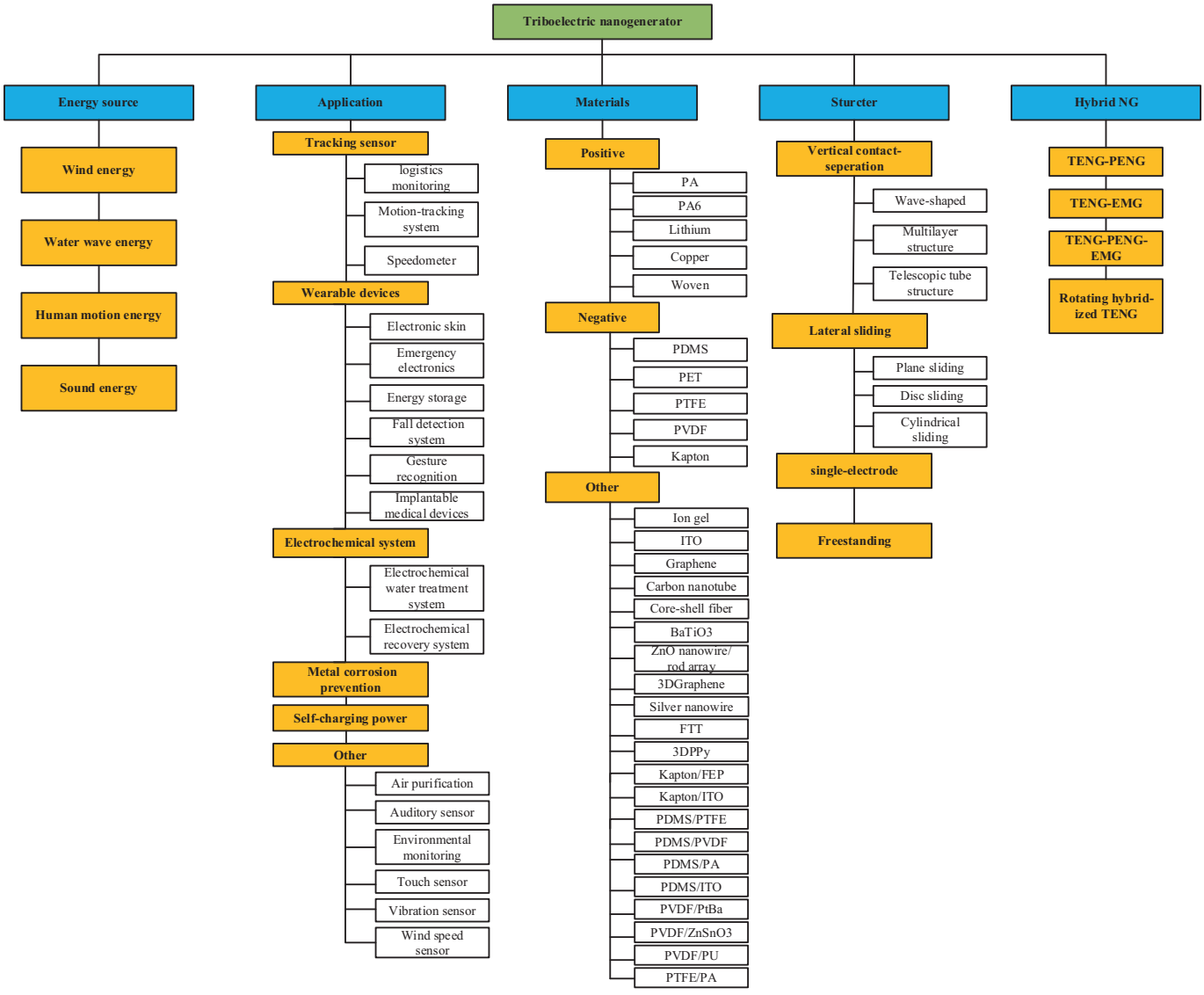


Fig. 6. The technology tree of TENG.

also some SAO structures that represent similar solutions information. By analyzing these SAO structures that represent similar solutions together, we can summarize a resolution for improving the output efficiency of TENG, that is, changing electrode materials. We also referred to this resolution as a second-level technical topic. That is, in Table 2 in the Appendix II, the second-level technical topic "Changing electrode materials" in the first-level technical topic E "Improving the output efficiency of TENG". In this paper, this type of second-level technical topic was referred to as technical resolution. Technical resolutions give different resolutions to the problem of improving efficiency of TENG.

In the first-level technical topics, we found that the relationship type represented by the SAO structures contained in topic B “Types of energy harvested by TENG”, topic G “Different materials of TENG”, and topic H “Hybrid NG” are mostly inclusion relationship type. For example, the SAO structure "layers obtain Polydimethylsiloxane" in first-level technical topic G “Different materials of TENG", the action "acquire" connects the subject "layers" and the object "Polydimethylsiloxane" together. This SAO structure illustrates the inclusion of Polydimethylsiloxane in the layers of the TENG. In the first-level technical topic G "Different materials of TENG", there are some SAO structures that also represent the same material information. By analyzing these SAO structures representing the same material information together, we can summarize a constituent material of TENG, that is, Polydimethylsiloxane. We referred to this material as a second-level technical topic. That is, in Table 2 in the Appendix II, the second-level technical topic "Polydimethylsiloxane (PDMS)" in the first-level technical topic G “Different materials of TENG”. In this paper, this type of second-level technical topic was referred to as technical components solution. The technical components solutions address the issues of energy harvesting types and materials for TENG, and compositions for hybrid NG.

After we conducted in-depth analysis of the SAO structure within each first-level technical topic, combining with the technology tree of TENG, we obtained the second-level technical topic corresponding to each first-level technical topic of the TENG, i.e., different technical solutions, and the results are shown in Table 2 in the Appendix II. When we obtained the second-level technical topics, we compared the second-level technical topics in the scientific papers and patents year by year. During the comparison process, we found that some second-level topics appeared in the scientific papers and patents at the same time, while some second-level topics only appeared once in the scientific papers but never appeared in patents. And some second-level topics appeared in the patents after they appeared in the scientific papers. According to the definition of the technology opportunity in this paper, we considered

that if a second-level technical topic, that is, a technical solution, appeared in scientific paper for two years or more that does not exist in patent, then it was considered as a technology opportunity. For example, in the first-level technical topic B "Types of energy harvested by triboelectric nanogenerators", there were second-level technical topics such as the collection of mechanical energy in the whole space, the collection of wind energy, the collection of water wave energy, the collection of human motion energy, and the collection of rotational energy. Among them, the second-level technical topic of collecting human movement energy appeared in scientific papers in 2014 and 2015, but it did not appear in the patent at the same year. We considered that the collecting human movement energy was a technology opportunity after 2015. In 2016, the collecting human movement energy appeared in the patent. This proves that the identification of technology opportunity was effective and feasible.

In all the second-level technical topics, there is one exception, topic A "Theoretical research of TENG" appeared in the scientific paper in 2012. The research under this topic was all about the theories of the triboelectric nanogenerator, such as the working principle of the triboelectric nanogenerator, the dielectric constant and work function and the morphological effect of charge transfer. Due to patents focus on the application of technology, the topic A did not appear in the patent in the following years. But we did not consider the second-level technical topics in the topic A as technology opportunity. Because, in fact, some of the theories in topic A were applied to technology development in the following years. For example, in 2015, some scholars proposed a hybrid nanogenerator model based on the working principle of the triboelectric nanogenerator. One year later, a cluster of hybrid nanogenerators appeared in scientific papers and patents, and the technology topic became a hot topic in the following years.

It should also be noted that the first-level technical topic I "Energy storage of TENG" shown in Table 2 disappeared after 2017. In fact, this topic has always existed, but since 2018, it has become the second-level topic "Energy collecting and storage integration" in the first-level topic F "wearable TENG", as shown in Table 2 in the Appendix II. As wearable devices have gradually become an important research direction, the problem of power shortage of wearable devices has gradually been paid attention to by researchers. Therefore, the integration of wearable devices with the energy collecting and storage of triboelectric nanogenerators has gradually become a hot research topic. This also reflects the advantages of SAO semantic mining, that is, it can directly identify technical problems and solutions in text information based on semantic relationship, and make up for the shortcomings of keywords that cannot express semantic information.

By analyzing the gaps between the second-level technical topics of scientific papers and of patents, we can identify potential technology opportunities at the technical solution level. For example, in the first-level topic C of the scientific paper, the second-level technical topic "A self-powered positioning and motion-tracking system based on triboelectric nanogenerator" appeared in scientific papers in 2017 and 2018, but it didn't appear in patents. The second-level technical topic " A self-powered positioning and motion-tracking system based on triboelectric nanogenerator" was considered as a potential technology opportunity based on the definition of technology opportunity in this paper. Table 3 depicts the potential technology opportunities obtained from the gaps between the second-level technical topics of scientific papers and of patents.

In terms of self-powered applications of TENG, Self-powered metal corrosion prevention system based on TENG, Self-powered positioning and motion-tracking system based on triboelectric nanogenerator and implantable medical devices were considered as potential technology opportunities. The corrosion is more likely to take place for metals under the ocean environment condition, the traditional cathodic protection needs either a sacrificial anode or an external power source, while TENG can solve the problem by converting the water wave energy to electricity (Li et al., 2015). Motion-tracking and positioning technologies promote a wide range of potential applications in smart interaction systems (Guo et al., 2018). Implantable medical devices are self-powered by capturing energy generated by human movement, muscle contraction/relaxation, heart/pulmonary movement and blood circulation to monitor biological signals and treat various diseases in vivo and in vitro (Ou et al., 2017).

Table 3
The potential technology opportunities obtained by the analysis of gaps between second-level technical topics in the scientific papers and patents.

First -level topics	Gaps between second-level technical topics of papers and of patents	Time of topics' appearance in scientific papers
Topic C: Self-powered applications of TENG	A self-powered system based on triboelectric nanogenerators for metal corrosion prevention	2014, 2015
	A self-powered positioning and motion-tracking system based on triboelectric nanogenerator	2017, 2018
	Implantable medical devices	2016, 2017, 2018, 2019
Topic E: Improving the output efficiency of TENG	Hierarchical structure	2018, 2019
Topic F: Wearable TENG	Self-powered wearable/portable emergency electronics	2018, 2019
Topic G: Different materials of TENG	Polyamide-6 (PA6) films	2018, 2019
	BaTio3 nanoparticles	2018, 2019
Topic H:Hybrid NG	Triboelectric-electromagnetic hybrid nanogenerator;	2016, 2017, 2018, 2019
	Triboelectric-piezoelectric-electromagnetic hybrid nanogenerator	2018, 2019
	Rotating hybride nanogenerator	2018, 2019

In terms of improving the output efficiency of TENG, the use of layered structures was considered as potential technology opportunity. The layered structure increases the triboelectric area, resulting in greater output power (Pang et al., 2019).

In terms of wearable TENG, wearable emergency electronic devices were considered as promising technology opportunities. Wearable emergency electronic equipment can obtain a variety of energy from the human body, effectively power the emergency electronic equipment, and can be used in field exploration, travel, and rescue without power supply (Cui et al., 2020).

In terms of materials used in TENG, polyamide 6 (PA6) films and BaTiO3 nanoparticles were considered as potential technology opportunities. TENG based on phase inversion membranes of PVDF and polyamide-6 (PA6) or TENG based on phase inversion membranes of PVDF and BaTiO3 nanoparticles can significantly enhance its performance (Jang and Oh, 2018; Shi et al., 2019b).

In terms of hybrid nanogenerators, triboelectric-electromagnetic hybrid nanogenerators, triboelectric-electromagnetic-piezoelectric hybrid nanogenerators, and rotating hybrid nanogenerators were considered as potential technology opportunities.

4.3. Analysis of the scientific papers related to TENG using outlier detection method

From the Figs. 1 to 8 in the Appendix I, it can be seen that some points are scattered outside the clusters and become outlier points, and are not included in the scope of clustering topics analysis. But these points might contain new technologies or technology opportunities. Therefore, in order to identify complete technology opportunities, we used the scientific papers of TENG from 2012 to 2019 as the data resource, and used the outlier detection algorithms to detect the outlier points of scientific paper clusters each year. We firstly used KNN algorithm, LOF algorithm, and IF algorithm to simultaneously detect outlier points in scientific paper clusters each year. Then we regarded the intersection of the results of these three algorithms as robust outlier scientific paper. Finally, the topics of outlier scientific papers were

compared with the topics of the patents, and we considered the topics of the outlier scientific papers that have not yet appeared in the patents as technology opportunities. Since the amount of data in 2012 was too small for training, we did not perform outlier detection on the scientific paper data in 2012. Table 4 shows the topics of the outlier scientific papers that were finally detected in scientific papers from 2013 to 2019.

Taking 2015 as an example, and through continuous debugging of the outlier detection model, we set the proportion of outlier points to 1%. We compared the outlier detection results of the three algorithms and took the intersection to obtain the annual robust outlier points. Figs. 7 to 9 are the outlier point detection results of KNN, LOF, and IF for scientific papers in 2015 respectively. In Figs. 7 to 9, the number below the point is the paper number, the black solid points represent outlier points, the hollow points represent normal points, and the red dotted line represents the boundary between the normal point and the outlier point calculated by the algorithm. The white area inside the red dashed line is the area where the normal point is located, and the blue area outside the red dashed line is the area where the outlier point is located. The deeper the blue where the outlier point is, the stronger the abnormality of the point. The processing method of the scientific papers data in other years was the same as the paper data of 2015.

After we obtained the annual outlier points in scientific papers, we observed the horizontal and vertical development process of each outlier point, that is, its development process in scientific papers and in patents. For the analysis of the horizontal development, we observed whether the topic of the outlier point form a cluster in the scientific papers or not. For the analysis of the vertical development, we observed whether the topic of the outlier point appear in the topic of the patents or not.

Table 4
The topics of outlier points in scientific papers from 2013 to 2019.

Years	Paper numbers	The topics of outlier scientific papers
2013	P1751	Piezoelectric-triboelectric hybrid nanogenerator
	P1760	Rotary triboelectric nanogenerator
	P1778	Harvest energy from human body movement
2014	P1665	Use naturally replicated micro-nano structured materials
	P1685	Woven structured triboelectric nanogenerator for wearable devices
	P1695	Segmented and multi-layer integrated TENG
	P1725	TENG for collecting vibration energy in whole space
	P1731	Collect human sports energy
2015	P1524	A hybridized power panel
	P1531	TENG with 3D printed materials
	P1600	Energy storage of TENG.
	P1656	Large-scale batch manufacturing of TENG
2016	P1307	Piezoelectric-triboelectric hybrid nanogenerator
	P1404	Triboelectric-electromagnetic hybrid nanogenerator
	P1430	Biodegradable triboelectric nano generator
	P1433	Textile-based wearable TENG
	P1474	An alternating current electroluminescent (ACEL) device using coupled zinc sulfide powder
2017	P1048	TENG based on air-driven membrane structure
	P1110	Chemically functionalized natural cellulose materials for effective TENG development
	P1215	Clean and detachable wearable TENG
	P1263	Harvest underwater ultrasonic wave energy
	P1297	Use TENG to detect wind speed from all directions of the aircraft
2018	P649	TENG based on rolling motion of beads
	P697	Used for driver behavior monitoring
	P783	Self-powered virtual reality three-dimensional control sensor based on triboelectric mechanism
	P919	Elastomerhydrogel mixture as power supply
	P962	Triboelectric nano-generator based on magnetic induction telescopic spring steel belt
2019	P1	Surface modification method based on neutral beam to improving triboelectric properties
	P293	Series-connected triboelectric nanogenerators
	P307	Flame retardant and self-extinguishing triboelectric nano generator
	P373	A triboelectric nanogenerator based on waste tea leaves and packaging bags
	P387	Organicinorganic nanostructured materials are integrated into the hybrid nanogenerator
	P477	Novel triboelectric nanogenerator based on electrospun polyvinylidene fluoride nanofibers

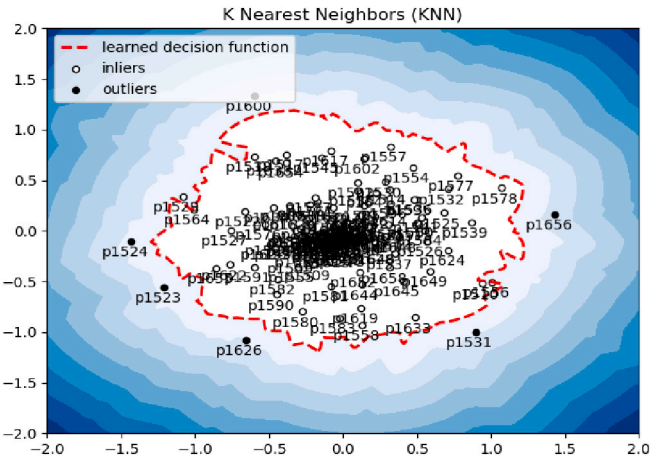


Fig. 7. Outlier detection based on KNN for scientific papers in 2015.

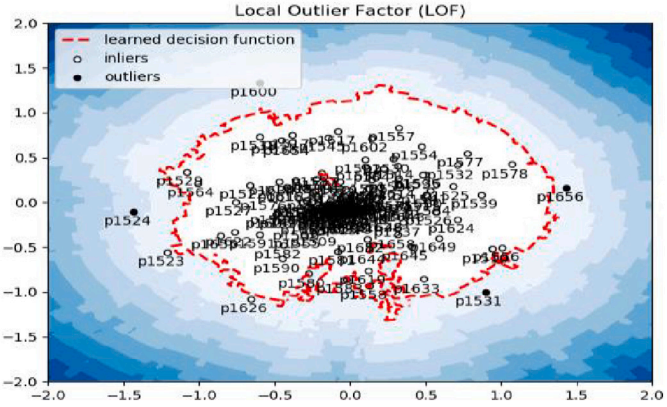


Fig. 8. Outlier detection based on LOF for scientific papers in 2015.

From the Table 2, Table 4 and the Table 2 in the Appendix II, we can find that the emergence and growth of TENG technology is traceable in scientific papers and patents. We found that some outlier points in scientific papers have developed into new technical solutions to existing technical problems. For example, the outlier scientific paper P1778 appeared in 2013, which described the TENG technology collecting motion energy from the human body. One year later, the topic of the outlier scientific paper formed a second-level topic cluster in the first-level technical topic B "Types of energy harvested by TENG". And in the 2016 patent, a technical solution for human body movement energy appeared. That is, the technical solution "collecting motion energy from the human body" first appeared in scientific papers in 2013, and then in 2016, this technical solution to the technical problem B "Types of energy harvested by TENG" appeared in the patent. Similarly, the outlier scientific paper p1433 appeared in 2016, and it was about a textile-based TENG. In 2017, it appeared in the first-level technical topic G "Different materials of TENG" as a second-level topic. In 2019, the textile-based TENG also appeared in the patent. That is, the technical solution "textile-based TENG" first appeared in scientific paper in 2016, and then in 2019, the technical solution to the technical problem G "Different materials of TENG" appeared in patent. In addition, the outlier scientific paper P1751 appeared in 2013, which described a piezoelectric-triboelectric hybrid nanogenerator. Three years later, in

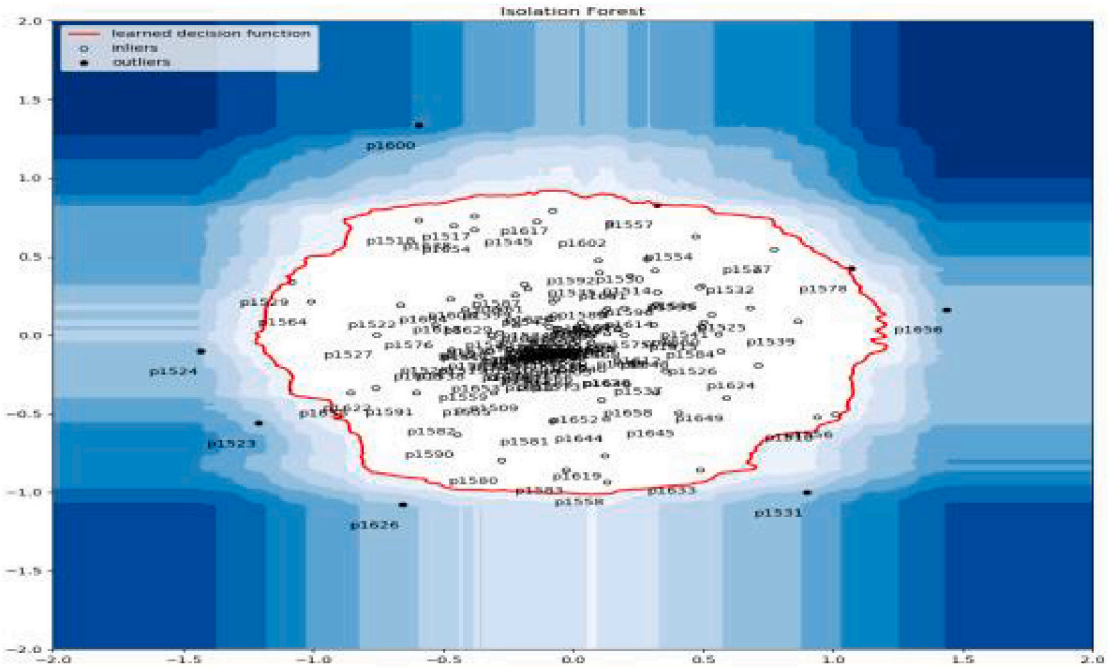


Fig. 9. Outlier detection based on IF for scientific papers in 2015.

2016, the topic of the outlier scientific paper formed a second-level topic in the first-level technical topic H "Hybrid NG". In the same year, a piezoelectric-triboelectric hybrid nanogenerator also appeared in the second-level topic of the patent, and the second-level topic continued to appear in the following years.

At the same time, we also found that some outlier points in scientific papers can be developed into a new technology, that is, a new first-level technical topic. For example, the outlier scientific paper P1307 appeared in 2014, which described a woven-structure triboelectric nanogenerator for wearable devices. One year later, in 2015, the stretchable and shape-adaptive wearable TENG formed a small cluster and appeared in scientific papers for the first time as the first-level technical topic F "wearable TENG". In 2016, the first-level technical topic F appeared in patents for the first time. Between 2016 and 2019, the number of scientific papers and patents on the topic F "wearable TENG" has increased rapidly, and its second-level topics have also become more abundant, such as stretchable and shape-adaptive wearable TENG, TENG for implantable medical devices, and wearable TENG with high durability and water resistance. Until now, the topic F has become a research focus and one of the research hotspots of TENG.

These results verify the rationality and feasibility of our attempts to find technology opportunities by the analysis of gaps between the topics of outlier scientific papers and of patents. Usually, a technology is different from the mainstream technology when it is first proposed, and only a few people conduct research. During the development of technology, more and more people pay attention to this technology, thus forming a research cluster. When basic research reaches a certain level of maturity, this research will be transformed into technology and applied. Therefore, when identifying technology opportunities, we should pay more attention to the outlier points in the clusters of scientific papers.

Finally, we compared the topics of the outlier scientific papers with the topics of patents, and considered the topics of outlier scientific papers that have not yet appeared in patents as potential technology opportunities. For instance, the topic in outlier scientific paper P1215 “Clean and detachable wearable TENG” appeared in 2017. It belongs to the first-level patent topic F “Wearable TENG”, but it has not appeared in the second-level patent topics since 2017. The topic “Clean and detachable wearable TENG” was considered as a potential technology opportunity according to the definition of technology opportunity in this paper. Table 5 displays the potential technology opportunities identified through the analysis of gaps between the topics of outlier scientific papers and of patents.

In terms of energy collecting of TENG, harvesting underwater ultrasonic energy was considered as a potential technology opportunity. Although the high pressure of water makes it difficult to obtain underwater acoustic energy, acoustic waves are a clean and sustainable energy source. The ocean contains a rich resource of ultrasound. Compared with traditional acoustic energy collection devices, the optimized TENG is more cost-effective, lighter weight and higher efficiency, it is an ideal choice for collecting ultrasonic energy underwater (Xi et al., 2017).

In terms of self-powered applications of TENG, large-scale batch manufacturing of TENG, use TENG to detect wind speed from all directions of the aircraft, driver behavior monitoring based on TENG, self-powered virtual reality three-dimensional control sensor based on triboelectric mechanism, flame retardant, and self-extinguishing triboelectric nanogenerator were considered as potential technology opportunities. The main parameters in driving behavior analysis are usually obtained by using some sensitive but expensive sensors, such as eye trackers. The triboelectric nanogenerator self-powered sensor has been proven to be feasible for monitoring driver behavior, and can greatly reduce the cost of driver behavior monitoring sensor devices (Meng et al., 2018).

In terms of improving the output efficiency of TENG, rotary TENG, segmented and multi-layer integrated TENG, the all-around series triboelectric nanogenerator with the microbead rolling structure, cascade impact structure, and surface modification method based on neutral beam were considered as potential technology opportunities. The resonant TENG produces a domino impact effect by hitting the next TENG. Due to the influence of the cascade, the non-resonant TENG will also produce output, which contributes to the expansion of the bandwidth (Bhatia et al., 2019).

In terms of wearable TENG, clean and detachable wearable TENG was considered as a potential technology opportunity.

In terms of different materials of TENG, naturally replicated micro-nano structured materials, ACEL equipment, air-driven membrane structures, hydrogel mixtures, biodegradable materials and electrospun polyvinylidene fluoride nanofibers, waste tea leaves and packaging

Table 5
The potential technology opportunities obtained by the analysis of gaps between the topics of outlier scientific papers and the topics of patents from 2013 to 2019

First-level topics	Gaps between the topics of outlier scientific papers and the topics of patents	Paper numbers	Time of topics' appearance in scientific papers
Topic B: Types of energy harvested by TENG	Harvest underwater ultrasonic wave energy	P1263	2017
Topic C: Self-powered applications of TENG	Large-scale batch manufacturing of TENG	P1656	2015
	Use TENG to detect wind speed from all directions of the aircraft	P1297	2017
	Driver behavior monitoring	P697	2018
	Self-powered virtual reality three-dimensional control sensor based on triboelectric mechanism	P783	2018
	Flame retardant and self-extinguishing triboelectric nano generator	P307	2019
Topic E: Improving the output efficiency of TENG	Rotary TENG	P1760	2013
	Segmented and multi-layer integrated TENG	P1695	2014
	Rolling motion of beads	P649	2018
	Cascade impact structure	P293	2019
	Surface modification method based on neutral beam to Improving triboelectric properties	P1	2019
Topic F: Wearable TENG	Clean and detachable wearable TENG	P1215	2017
Topic G: Different materials of TENG	Use naturally replicated micro-nano structured materials	P1665	2014
	Alternating current electroluminescence (ACEL) device	P1474	2016
	Air-driven membrane structures	P1048	2017
	Biodegradable materials	P1110	2017
	Hydrogel Hybrids	P919	2018
	Electrospun polyvinylidene fluoride nanofibers	P477	2019
	Waste tea leaves and packaging bags	P373	2019
	Organicinorganic nanostructured materials	P387	2019
Topic H:Hybrid NG	Hybrid solar panel integrated with TENG	P1524	2015
	Triboelectric-electromagnetic hybrid nanogenerator	P1404	2016

bags, and organic nanostructured materials were considered as potential technology opportunities. Especially the biodegradable materials as an outlier point consecutively appeared in 2016 and 2017, and it appeared in the second-level topic "biodegradable materials" in 2018 and 2019. But it has never appeared in the patents until 2019. From the perspective of the material itself, most TENGs use polymers as substrates or active layers, which causes plastic pollution (Khandelwal et al., 2019). The TENG using biodegradable materials can not only reduce resource waste, but also reduce environmental pollution and energy consumption (Xia et al., 2019). Therefore, in the field of TENG materials, biodegradable materials should be paid attention to by scholars.

In terms of Hybrid NG, Hybrid solar panel integrated with TENG and triboelectric-electromagnetic hybrid nanogenerator were considered as potential technology opportunities.

4.4. Identification of technology opportunities for TENG

From the above analysis, we can see that we have identified some potential technology opportunities by comparing the gaps between the second-level topics in scientific papers and the second-level topics of patents related to TENG, and by analyzing the gaps between the topics of outlier scientific papers and of patents. Through the analysis of the potential technology opportunities obtained above, we found that the potential technology opportunities were at the level of technical solutions. Based on the identified results of the above potential technology opportunities, in order to more accurately identify technology opportunities in the field of TENG, we needed to analyze the development trends of the first-level technical topics related to TENG. In another words, we needed to consider those technical problems with better development trends or higher growth rates. Because the high growth rate of certain technical topics indicates that they have a potential growth trend and they may get more attention from researchers after a few years (Bildosola et al., 2017; Li et al., 2019). The technical solutions corresponding to these technical topics will be more valuable and meaningful for enterprises' R&D decision-making.

Some first-level technical topics with low-profile and low-growth were not considered in the scope of our consideration of identifying technology opportunities. For example, the first-level technical topic D as shown in Table 1 exists in scientific papers and does not exist in patents. According to the definition of technology opportunity in this paper, it should be considered as technology opportunity. But the topic D disappeared after 2015. This shows that this technical topic is not being paid attention to. Therefore, when we finally identify the technology opportunities, we need to make judgments based on the development trend of technical problems and the trend of the technical problem's growth rate. We calculated the number of scientific papers and patents contained in the first-level topics corresponding to the potential technology opportunities obtained in Section 4.2 and 4.3, as well as the growth rate. The results are shown in Figs. 10-15 respectively.

From Fig. 10 to Fig. 15, we can see that the number of scientific papers and patents contained in these six first-level technical topics has shown a steady growth trend, and the topic's growth rate has also shown an increasing trend. This shows that these six technical topics have always been the focus of attention of researchers and developers of TENG. Therefore, the technical solutions corresponding to these six technical problems may be of great interest and concern to R&D experts.

To obtain the final technology opportunities, based on the potential technology opportunities results obtained in Table 3, Table 5, and the first-level technical topic's growth rate results depicted in Figs. 10 to 15, we invited two domain experts to screen the technology opportunities from the three aspects of technological development potential, technological feasibility, and economic feasibility. Finally, we identified the final technology opportunities related to TENG after 2019, as shown in Table 6.

5. Conclusions

With the rapid advancement of science and technology, how to early identify technology opportunity in the gaps between science and technology has become a critical task for R&D managers and policy makers. This study attempted to advance the technology opportunity research field forward by combining information from both scientific papers and patents, as well as integrating SAO semantic mining and outlier detection method. To do so, this paper proposed a research framework that utilized scientific papers and patents as data resources and integrated SAO semantic mining and outlier detection method to identify technology opportunities. The triboelectric nanogenerator technology was selected as a case study to demonstrate the validity and flexibility of the proposed framework. The framework provides a tool for understanding the evolutionary paths of science and technology, and for identifying technology opportunity by analyzing the gaps between science and technology.

In previous studies on identifying technology opportunities by analyzing the gaps between science and technology, scholars have

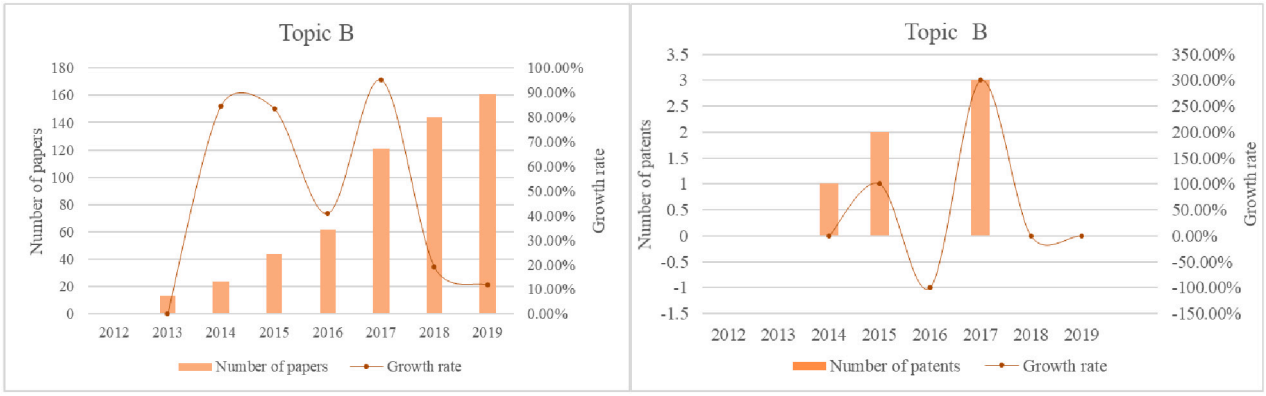


Fig. 10. The number of scientific papers and patents contained in topic B and the growth rate from 2012 to 2019.



Fig. 11. The number of scientific papers and patents contained in topic C and the growth rate from 2012 to 2019.

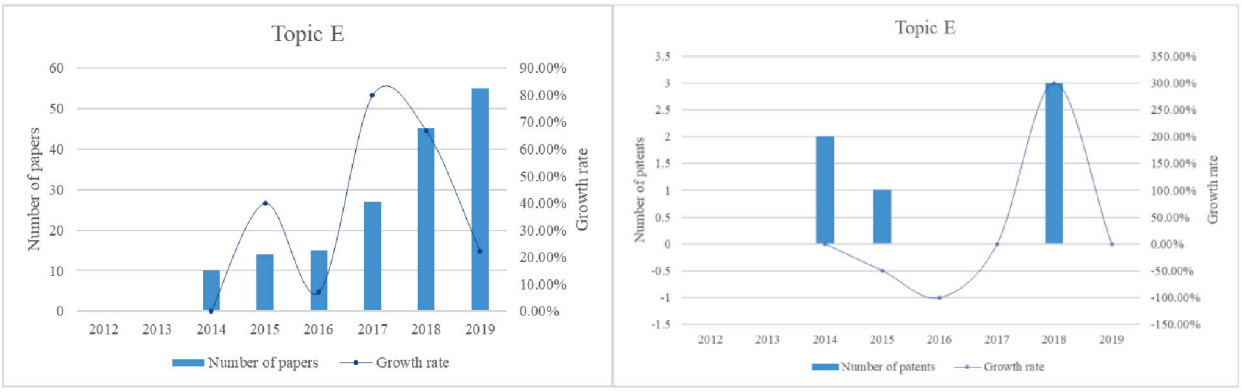


Fig. 12. The number of scientific papers and patents contained in topic E and the growth rate from 2012 to 2019.

focused more on the difference between technical topics' first appearance in scientific papers and in patents. However, few scholars conduct in-depth analysis of the detailed information and evolution of technical topics in scientific papers and patents. Therefore, in this paper, we tried to analyze the details of the technical topics in scientific papers and patents, and analyze the difference between the evolution of technical topics in scientific papers and patents in order to deeply analyze the gaps between science and technology. We firstly divided technical topics in scientific papers and patents into first-level and second-level technical topics. Then we monitored the evolution of first-level and second-level technical topics in scientific papers and patents, especially the emergence and growth of technical topics in outlier scientific papers. Finally, we mainly analyzed two different gaps between science and technology: One gap shows the difference between first-level and second-level technical topics' first appearance in scientific papers and in patents, and the other is the differences analysis between the time of topics' appearance in outlier scientific papers and the time when the topics' first appearance in patents. Some key findings and contributions are listed as follows:

Firstly, in previous studies on identifying technology opportunities by analyzing the gaps between science and technology, most scholars only analyze the main contents of the clusters of scientific papers and patents. Outlier points scattered outside the clusters are frequently ignored. Some researchers only focus on studying the outliers, but do not consider clustering information. These approaches produce insufficient results for identifying technology opportunities. In order to identify

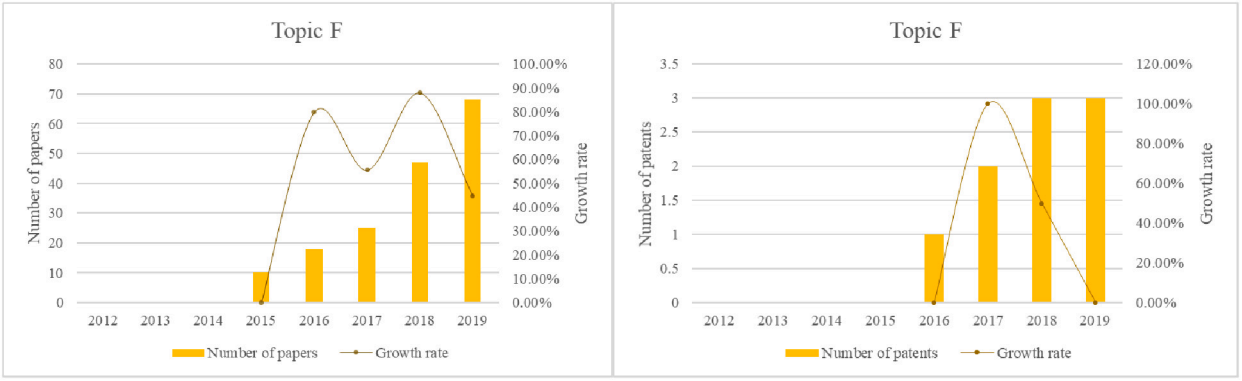


Fig. 13. The number of scientific papers and patents contained in topic F and the growth rate from 2012 to 2019.

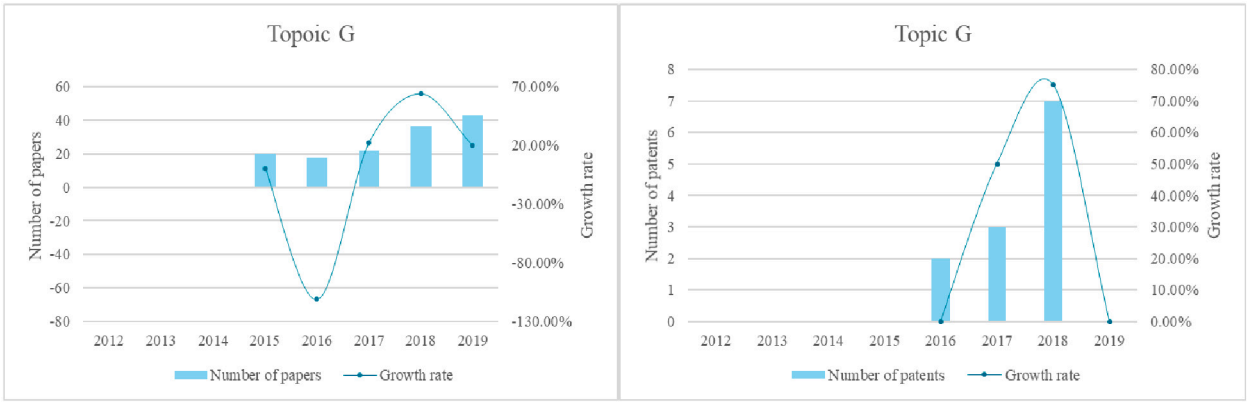


Fig. 14. The number of scientific papers and patents contained in topic G and the growth rate from 2012 to 2019.

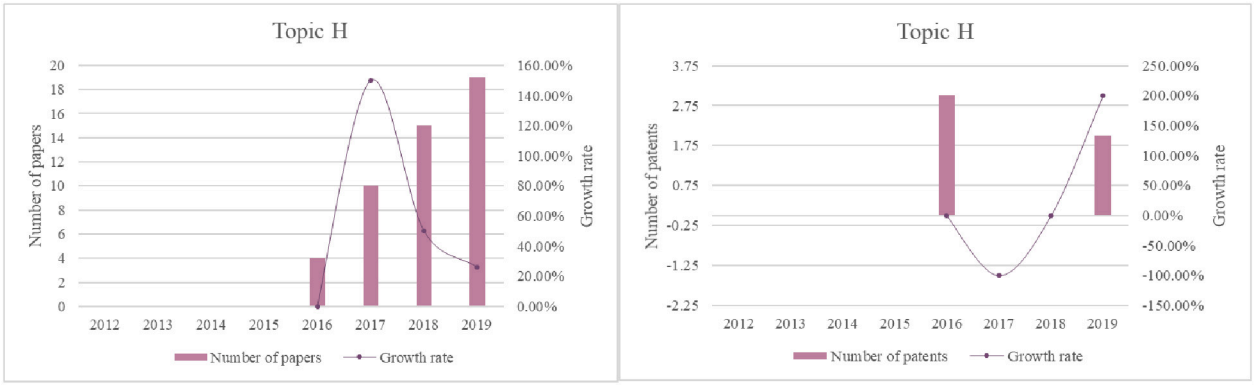


Fig. 15. The number of scientific papers and patents contained in topic H and the growth rate from 2012 to 2019.

complete technology opportunities, in this framework, we incorporated clustering and outlier points information into the scope of technology opportunities identification. We not only analyzed the technical topics obtained from clusters of scientific papers and patents, but we also used three different outlier detection algorithms to detect robust outlier points in scientific papers. And we identified potential technology opportunities by the analysis of gaps between the topics of outlier scientific papers and of patents. We also tracked and monitored the emergence and growth of outlier points. We found that some outlier points in scientific papers have developed into new technical solutions to existing technical problems, and we also found that some outlier points have developed into a new technology, i.e., a new first-level technical topic. These findings may have important implications for our understanding of the evolutionary paths of science and technology, as well as for identifying technology opportunities in the gaps between science and technology.

Secondly, many scholars have used SAO semantic mining to identify technology opportunities. However, they only used the SAO semantic analysis method for text clustering analysis. The in-depth semantic analysis of the SAO structure was not performed, resulting in the identified technology opportunities with a higher granularity. In this paper, we deeply analyzed the technical information represented by SAO structures extracted from scientific papers and patents. We mainly focused on the information of technology-function relationship, problem-resolution relationship, and inclusion relationship among technical components represented by SAO structures. In the process of

Table 6
Technology opportunities related to TENG after 2019

First-level topics	Technology opportunities	Resources	Time of topics' appearance in scientific papers
Topic B: Types of energy harvested by TENG	Harvest underwater ultrasonic wave energy	Gaps between the topics of outlier scientific papers and the topics of patents	2017
Topic C: Self-powered applications of TENG	Motion tracking systems based on the self-powered TENG	Gaps between second-level technical topics of papers and patents	2017, 2018
	Implantable medical devices	Gaps between second-level technical topics of papers and patents	2016, 2017, 2018, 2019
Topic E: Improving the output efficiency of TENG	Hierarchical structure	Gaps between second-level technical topics of papers and patents	2018, 2019
	Rolling motion of beads	Gaps between the topics of outlier scientific papers and the topics of patents	2018
	Cascade impact structure	Gaps between the topics of outlier scientific papers and the topics of patents	2019
Topic F: Wearable TENG	Emergency electronics	Gaps between second-level technical topics of papers and patents	2018, 2019
Topic G: Different materials of TENG	BaTio3 nanoparticles	Gaps between second-level technical topics of papers and patents	2018, 2019
	Biodegradable materials	Gaps between the topics of outlier scientific papers and the topics of patents	2017
	Hydrogel Hybrids	Gaps between the topics of outlier scientific papers and the topics of patents	2018
	Electrospun polyvinylidene fluoride nanofibers	Gaps between the topics of outlier scientific papers and the topics of patents	2019
Topic H:Hybrid NG	Triboelectric-electromagnetic hybrid nanogenerators	Gaps between second-level technical topics of papers and patents; Gaps between the topics of outlier scientific papers and the topics of patents	2016, 2017, 2018, 2019
	Triboelectric-electromagnetic-piezoelectric hybrid nanogenerators	Gaps between second-level technical topics of papers and patents	2018, 2019
	Rotating hybrid nanogenerators	Gaps between second-level technical topics of papers and patents	2018, 2019

In the first-level technical topic B: Harvesting underwater ultrasonic wave energy would be a technology opportunity.
In the first-level technical topic C: Self-powered positioning and motion-tracking system based on triboelectric nanogenerator and implantable medical devices would be technology opportunities.
In the first-level technical topic E: the use of layered structure, bead rolling structure, series structure of triboelectric nanogenerator with cascade impact and the use of hybrid nanogenerator to improving output efficiency would be technology opportunities.
In the first-level technical topic F: wearable emergency electronic devices would be promising technology opportunities.
In the first level technical topic G: BaTiO3 nanoparticles, electrospun polyvinylidene fluoride, biodegradable materials, and hydrogel mixtures as electrodes would be technology opportunities. The materials used in TENG would develop in the direction of environmental protection and reduction of energy consumption.
In the first-level technical topic H: triboelectric-electromagnetic hybrid nanogenerators, triboelectric-electromagnetic-piezoelectric hybrid nanogenerators, and rotating hybrid nanogenerators would be technology opportunities. Reducing the waste of energy would be the main development direction of hybrid nanogenerators.

analyzing the clustering topics of SAO structures, we found that we can divide the technical information represented by SAO structures into first-level technical topics (technical problems) and second-level technical topics (technical solutions). By comparing the difference between first-level topics' and second-level topics' first appearance in scientific papers and patents, we found that we can identify potential technology opportunities not only at the level of technical problems, but also at the level of technical solutions. The technical problems are new technical trends or new technologies. The technical solutions are solutions to existing technical problems. These findings contribute to our understanding of the application of SAO semantic analysis methods. At the same time, these findings also broaden our understanding of identifying technology opportunities at various levels by analyzing the gaps between science and technology. The identified technology opportunities at the level of technical problems and technical solutions will serve as important decision-making references for R&D managers and policymakers. Therefore, this framework improves the completeness of the results of technology opportunities identification, while also providing important methodological support for government R&D strategic planning, social investment, and enterprise R&D decision-making.

However, this study still has several limitations. First, we only used the WOS database and DI database as data resources. Future research will consider more data from market information or user needs to identify technology opportunities among basic research, applied research, and market demand. Second, in this paper, we only analyzed the three types of technical relationships represented by the SAO structure. In the future, we will consider richer semantic information represented by the SAO structure to realize more in-depth text information mining for technology opportunities identification.

Data availability

Data will be made available on request.

Acknowledgments

This paper is supported by the National Natural Science Foundation of China (Grant 72174017, 71673018). Qianqian Xie is financially supported by the China Scholarship Council (Grant No. 201906540033). We are grateful to the two anonymous reviewers for their valuable comments.

Appendix I

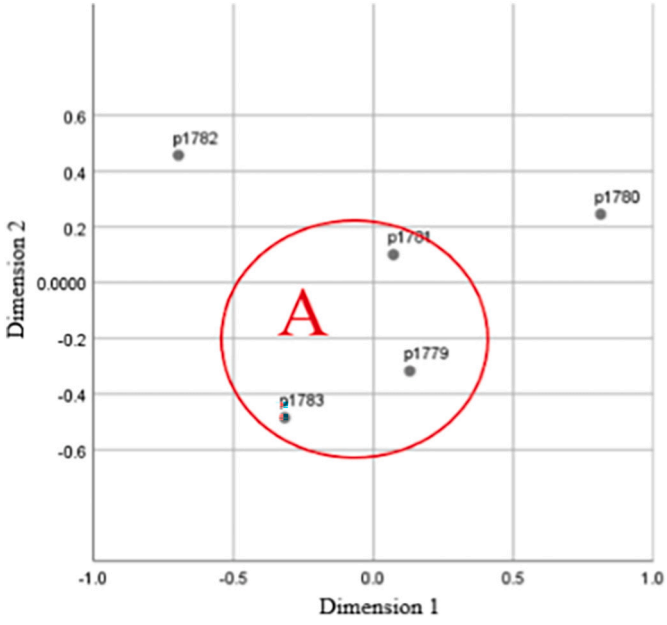


Fig. 1. Cluster diagram of scientific papers in 2012.

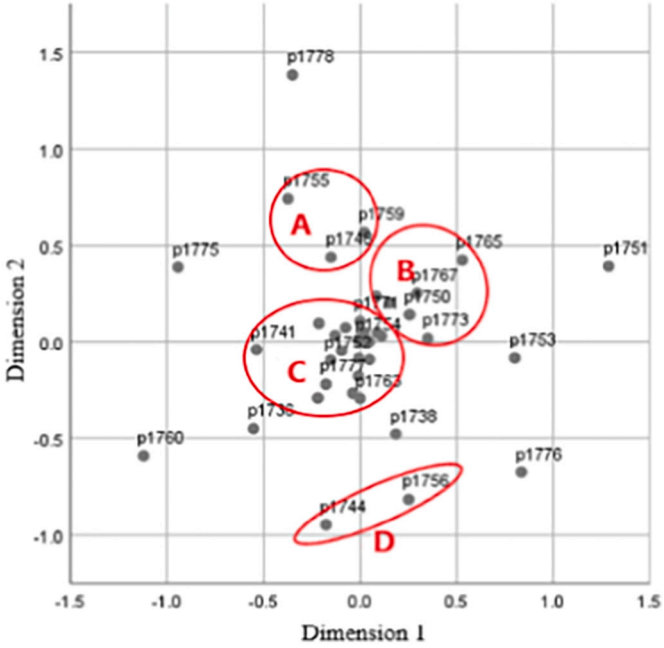


Fig. 2. Cluster diagram of scientific papers in 2013.

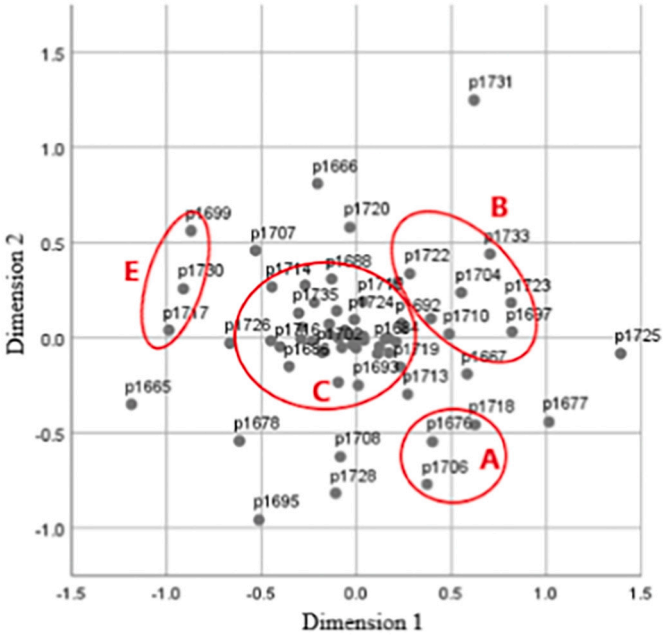


Fig. 3. Cluster diagram of scientific papers in 2014.

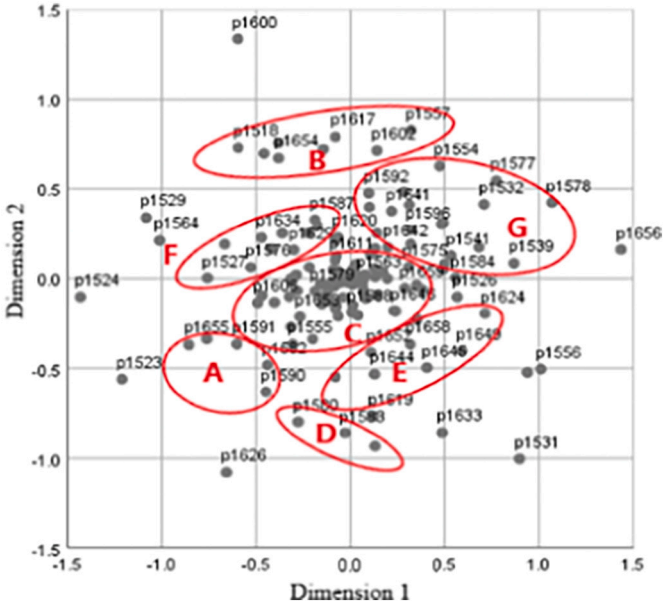


Fig. 4. Cluster diagram of scientific papers in 2015.

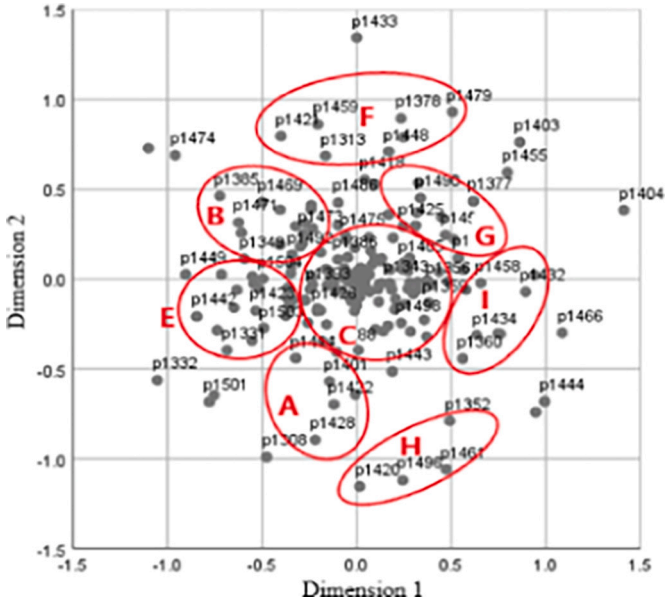


Fig. 5. Cluster diagram of scientific papers in 2016.

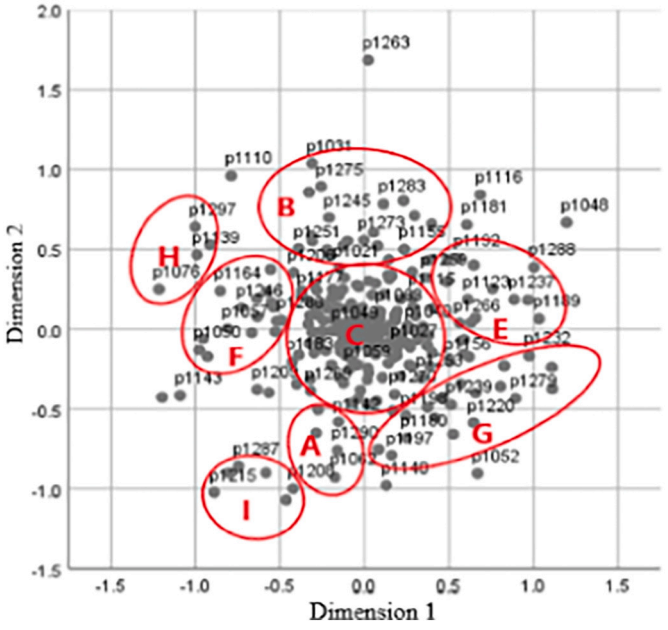


Fig. 6. Cluster diagram of scientific papers in 2017.

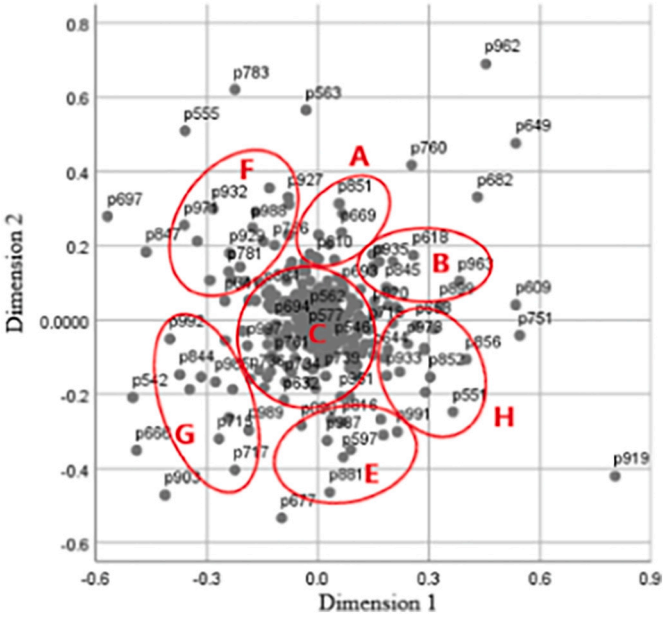


Fig. 7. Cluster diagram of scientific papers in 2018.

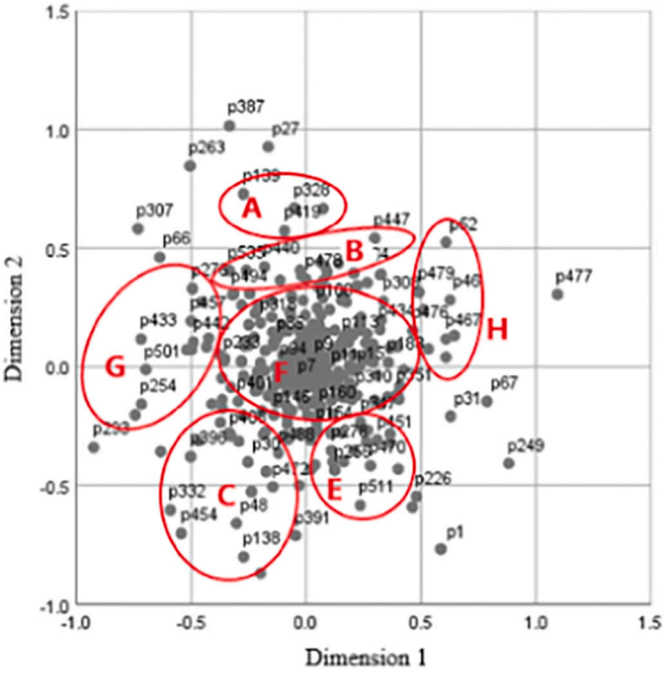


Fig. 8. Cluster diagram of scientific papers in 2019.

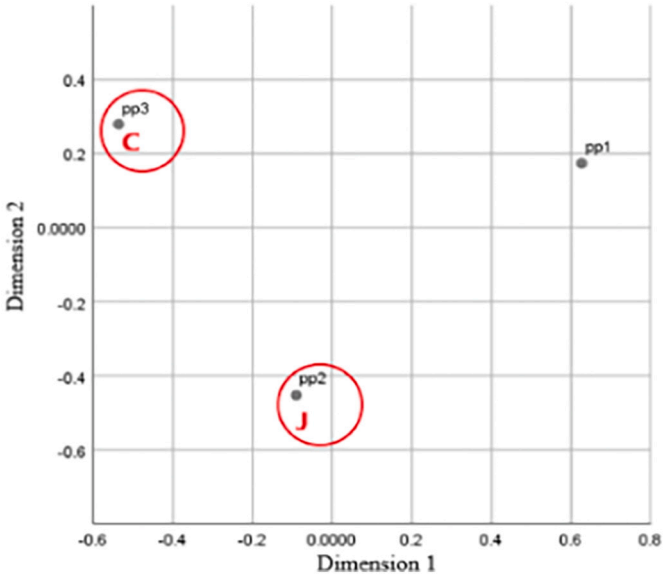


Fig. 9. Cluster diagram of patents in 2012.

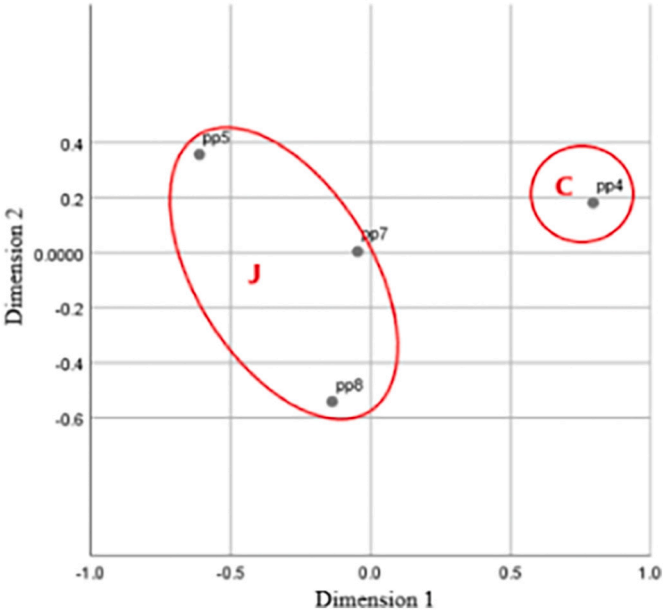


Fig. 10. Cluster diagram of patents in 2013.

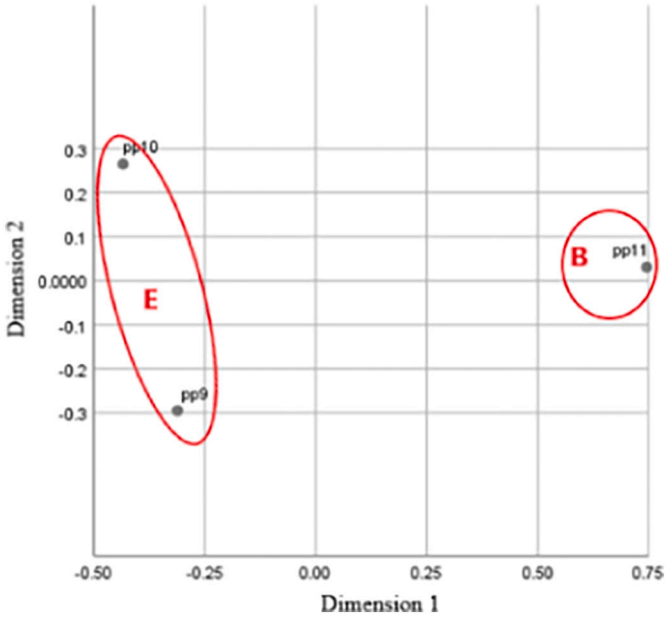


Fig. 11. Cluster diagram of patents in 2014.

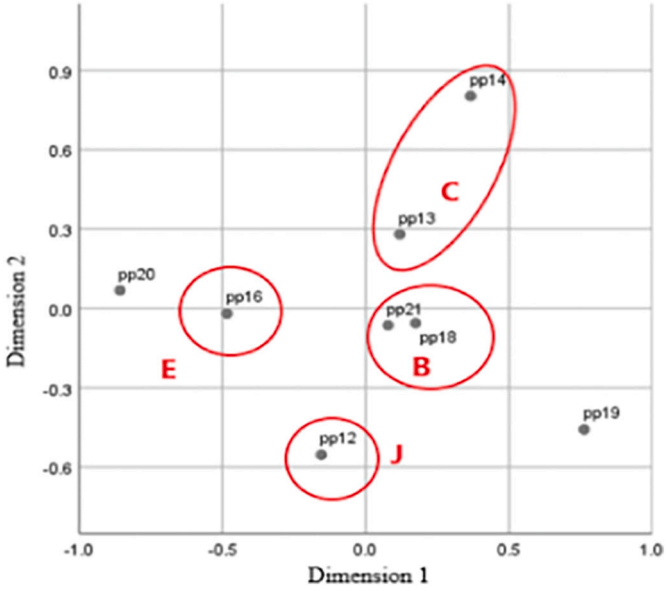


Fig. 12. Cluster diagram of patents in 2015.

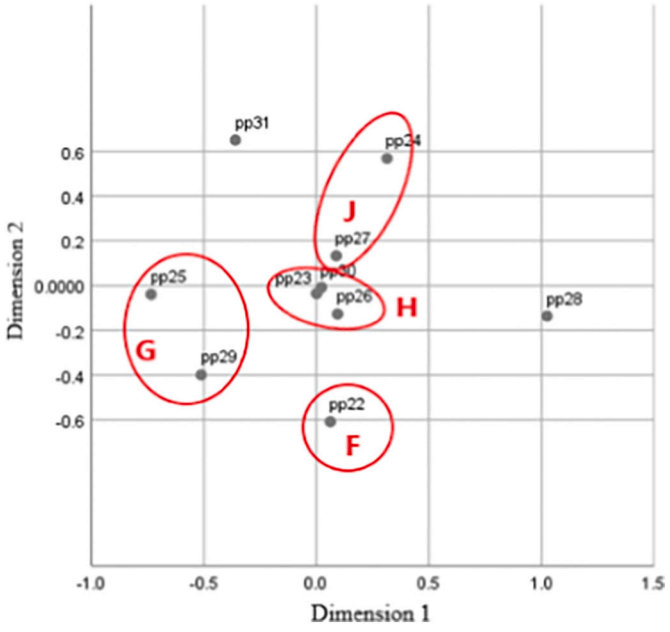


Fig. 13. Cluster diagram of patents in 2016.

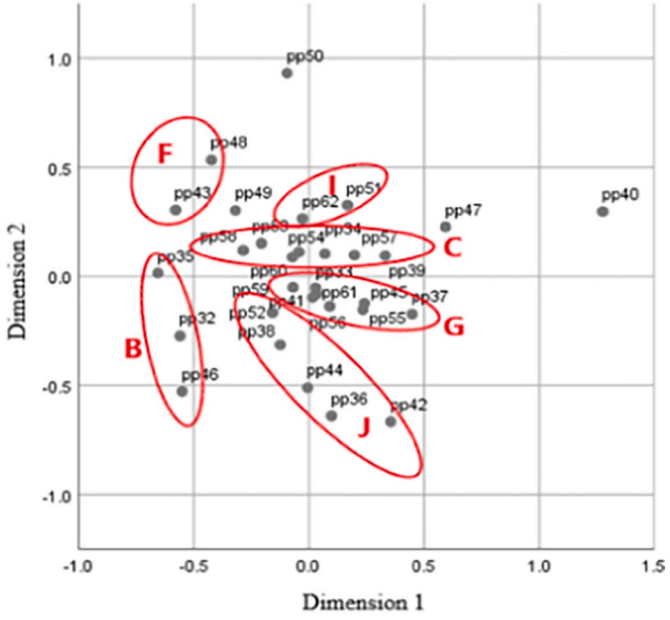


Fig. 14. Cluster diagram of patents in 2017.

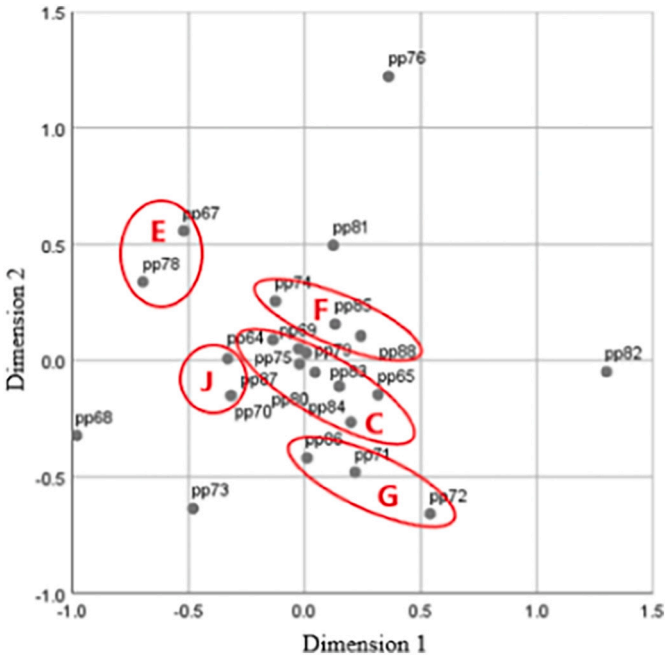


Fig. 15. Cluster diagram of patents in 2018.

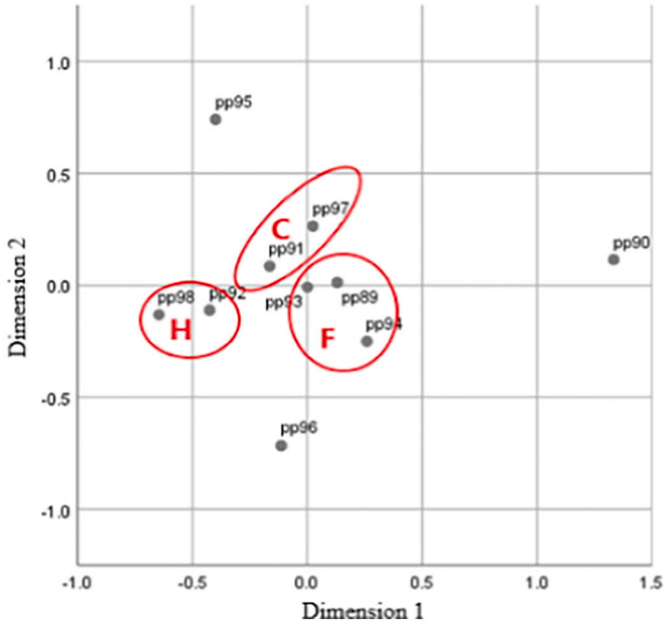


Fig. 16. Cluster diagram of patents in 2019.

Appendix II

Table 1
Short name of the topic related to triboelectric nanogenerator.

Full name	Short name
Subject-Action-Object	SAO
Triboelectric nanogenerator	TENG
Piezoelectric nanogenerator	PENG
Electromagnetic generator	EMG
Nanogenerator	NG
Polydimethylsiloxane	PDMS
Polyvinylidene difluoride	PVDF
Poly tetra fluoroethylene	PTEF

(*continued on next page*)

Table 1 (*continued*)

Full name	Short name
Polyethylene terephthalate	PET
Polyamide-6	PA6
Three dimension	3D
Three-dimensional graphene	3DG
Multi-dimensional scaling	MDS

Table 2
Comparison of second-level technical topics in scientific papers and patents.

First-level technical topics	Data source	2012	2013	2014	2015	2016	2017	2018	2019
A: Theoretical research of TENG	scientific papers	Working principle	Working principle	Working principle; materials performance	Working principle; Logical operation and mechanical sensing of triboelectronic devices; Output power; Hybrid nanogenerator	Dielectric constant and work function; Flutter phenomenon in flow-driven energy harvester; Triboelectric potential gate;	Morphologica effect of charge transfer; TENG's potential in Mars exploration	Working principle; Unified theoretical model applicable to different types of triboelectric nanogenerators; potential of liquid sensor; Literature Review of Triboelectric Nano-generator	Effective maximum energy output and modified quality factor; Literature review of TENG
B: Types of energy harvested by TENG	scientific papers	\	Mechanical energy	Mechanical energy; Acoustic energy; Wind energy; Water wave energy; Human body movement energy; Rotational energy	Harvest mechanical energy in full space; Acoustic energy; Wind energy; Water wave energy; Human body movement energy	Harvest mechanical energy in the whole space; Wind energy; Water wave energy; Human body movement energy;	Harvest mechanical energy in the whole space; Wind energy; Water wave energy; Human body movement energy; Heat energy	Harvest mechanical energy in the whole space; Wind energy; Water wave energy; Human body movement energy; Rotational energy	Harvest mechanical energy in the whole space; Wind energy; Water wave energy; Human body movement energy; Rotational energy
	Patents	\	\	Water wave energy; Rotational energy	Wind energy; Water wave energy	Human body movement energy	Mechanical energy; Water wave wind energy; Human body movement energy	Harvest mechanical energy in all directions; solar energy power; wind energy; Human body movement energy	Mechanical energy; Water wave energy
C: Self-powered applications of TENG	scientific papers	\	Touch sensor	Metal anticorrosion; Self-powered tracking system; Logistics control; Self-powered page marking and anti-theft sensor; Self-powered photodetector	Metal anticorrosion Self-charging power supply; Air purification system; Alcohol breath analyzer; Electrochemical water treatment system	Implantable medical equipment; Self-charging power system; Organic lead halide perovskite single crystal photodetector; Electrochemical recovery system; Vibration sensor; Honey detection method	Motion tracking system; Implantable medical equipment; Detecting the environmental humidity; Fall detection system; Electronic skin	Motion tracking system; Gesture recognition; Biomedical applications; Rotating state monitoring; Hearing sensor; Liquid level fluctuation sensor; Implantable medical equipment	Gesture recognition; Biomedical applications; Electrochemical system; Tire pressure monitor; Respiratory detection system; Implantable medical equipment
	Patents	Collect electric energy from sole	Wine bottle anti-counterfeit device	\	Self-charging for mobile electronic equipment; Motion sensor; Electrochemical application system;	\	Used in energy systems; Capacitance sensor; Wind speed sensor	Biomedical applications; Bionic manipulator; Sensing system of shoes;	Immune detection; Multimedia velocimeter; Intelligent jigsaw puzzle; Wireless optical communication system; Inventory management system
D: Integration of TENG with other batteries for mixed energy Battery	scientific papers	\	Solar cells and TENG form a hybrid energy cell	\	Solar cells and TENG form a hybrid energy cell	\	\	\	\

(*continued on next page*)

Table 2 (*continued*)

First-level technical topics	Data source	2012	2013	2014	2015	2016	2017	2018	2019
E: Improving the output efficiency of TENG	scientific papers	\	\	Changing electrode materials; Use sliding triboelectric method	Enhance the effective contact area of two materials with opposite polarities of triboelectric	Two generator devices are connected in parallel; Electric double layer effect; Bipiezoelectric driving mechanism; Surface microstructure modification method;	Use hybrid nanogenerators; Control the dielectric constant	Adopt hierarchical structure; Use hybrid nanogenerators Introducing spring structure; Use materials with low thermal conductivity; Surface microstructure modification method;	Adopt hierarchical structure; Use hybrid nanogenerators; Changing electrode materials; Use an external charge pump;
	Patents	\	\	Use telescopic tube structure; Use sliding triboelectric method	Changing electrode materials; Use wavy triboelectric structure;	\	\	The surface of the contact layer protrudes	\
F: Wearable TENG	scientific papers	\	\	\	Stretchable and shape adaptive	Stretchable and shape adaptive; Wearable TENG in harsh environment	Stretchable and shape adaptive; Portability; Power shortage of wearable devices;	Stretchable and shape adaptive; Portability; Emergency rescue; High durability and water resistance; Energy collection and storage integration;	Stretchable and shape adaptive; Portability; Emergency rescue; Cleanable; Energy collection and storage integration
	Patents	\	\	\	\	Stretchable and shape adaptive; Wear resistance;	Stretchable and shape adaptive; Portability; Piezoelectric triboelectric hybrid wearable nano generator; Energy storage of wearable TENG	Stretchable and shape adaptive; Portability;	Stretchable and shape adaptive; Portability;
G: Different materials of TENG	scientific papers	\	\	\	Polydimethylsiloxane (PDMS); Polyethylene terephthalate(PET); Graphene; Multi-walled carbon nanotubes;	PDMS; PET; Graphene; Multi-walled carbon nanotubes; Polyvinylidene fluoride (PVDF); Lithium ion battery; Flexible triboelectric transistor; Other materials and PVDF;	PDMS; PET; Graphene; Multi-walled carbon nanotubes; PVDF ; Flexible triboelectric transistor; Textiles; Barium titanate nanoparticles; Other materials and polymers of PVDF; Ionic gel; Flexible electrospun paper	PDMS; PET; Graphene; Multi-walled carbon nanotubes; PVDF ; Lithium ion battery; Textiles; 3D printed materials; Silver nanowires; Polyamide -6 (PA6) film; Barium titanate nanoparticles; Ionic gel; Other materials and polymers of PDMS; Polytetrafluoroethylene (PTEF)	PDMS; PET; Three-dimensional graphene (3DG) film; Multi-walled carbon nanotubes; PVDF ; Lithium ion battery; Textiles; 3D printed materials; Silver nanowires; Polyamide -6 (PA6) film; Barium titanate nanoparticles; Other materials and polymers of PDMS; Other materials and polymers of PVDF; Other materials and polymers of PTEF; Biodegradable materials
	Patents	\	\	\	\	PDMS ; PET; graphene	PDMS ; PET; Graphene; PVDF ; Ionic gel; Other materials	PDMS; PET; Graphene; PVDF ; Silver nanowires; Other materials and	PDMS; PET; Graphene; 3D printed materials; Multi-walled

(*continued on next page*)

Table 2 (*continued*)

First-level technical topics	Data source	2012	2013	2014	2015	2016	2017	2018	2019
							and polymers of polyvinylidene fluoride (PVDF); Paper-based composites; Chitosan	polymers of PDMS; Flexible triboelectric transistor; Compressing wood substrate	carbon nanotubes; Textiles; Ionic gel
H: Hybrid NG	scientific papers	\	\	\	\	Triboelectric-electromagnetic hybrid nanogenerator; Piezoelectric-triboelectric hybrid nanogenerator; Triboelectric generators and thermoelectric generator;	Triboelectric-electromagnetic hybrid nanogenerator; Piezoelectric-triboelectric hybrid nanogenerator	Triboelectric-electromagnetic hybrid nanogenerator; Triboelectric-piezoelectric-electromagnetic hybrid nanogenerator; Rotating hybrid triboelectric nanogenerator; Piezoelectric-triboelectric hybrid nanogenerator	Triboelectric-electromagnetic hybrid nanogenerator; Triboelectric-piezoelectric-electromagnetic hybrid nanogenerator; Rotating hybrid triboelectric nano-generator; Piezoelectric-triboelectric hybrid nanogenerator; Paper-based hybrid nanogenerator
	Patents	\	\	\	\	Piezoelectric-triboelectric hybrid nanogenerator	Piezoelectric-triboelectric hybrid nanogenerator	Piezoelectric-triboelectric hybrid nanogenerator	Piezoelectric-triboelectric hybrid nanogenerator

References

Alghushairy, O., Alsini, R., Soule, T., Ma, X., 2020. A review of local outlier factor algorithms for outlier detection in big data streams. Big Data Cogn. Comput. 5 (1), 1–24.

Bhatia, D., Hwang, H.J., Huynh, N.D., Lee, S., Nam, Y., Kim, J., Choi, D., 2019. Continuous scavenging of broadband vibrations via omnipotent tandem triboelectric nanogenerators with cascade impact structure. Sci. Rep. 9 (1), 1–9.

Bildosola, I., Río-Bélver, R.M., Garechana, G., Cilleruelo, E., 2017. TeknoRoadmap: an approach for depicting emerging technologies. Technol. Forecast. Soc. Chang. 117, 25–37.

Breunig, M.M., Kriegel, H.P., Ng, R.T., Sander, J., 2000. LOF: identifying density-based local outliers. In: Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, pp. 93–104.

Cascini, G., Fantechi, A., Spinicci, E., 2004. Natural language processing of patents and technical documentation. In: International Workshop on Document Analysis Systems Springer, Berlin, Heidelberg, pp. 508–520.

Cho, J., Lee, J., 2013. Development of a new technology product evaluation model for assessing commercialization opportunities using Delphi method and fuzzy AHP approach. Expert Syst. Appl. 40 (13), 5314–5330.

Choi, S., Yoon, J., Kim, K., Lee, J.Y., Kim, C.H., 2011. SAO network analysis of patents for technology trends identification: a case study of polymer electrolyte membrane technology in proton exchange membrane fuel cells. Scientometrics 88 (3), 863–883.

Choi, J., Jeong, B., Yoon, J., 2019. Technology opportunity discovery under the dynamic change of focus technology fields: application of sequential pattern mining to patent classifications. Technol. Forecast. Soc. Chang. 148, 119737.

Cozzens, S., Gatchair, S., Kang, J., Kim, K.S., Lee, H.J., Ordóñez, G., Porter, A., 2010. Emerging technologies: quantitative identification and measurement. Tech. Anal. Strat. Manag. 22 (3), 361–376.

Cui, X.J., Cao, S.L., Guo, R., Khan, S.A., Xie, G., Tian, Z., Sang, S.B., Zhang, H.L., 2020. Building self-powered emergency electronics based on hybrid nanogenerators for field survival/rescue. Energy Sci. Eng. 8 (3), 574–581.

Dang, T.T., Ngan, H.Y., Liu, W., 2015. Distance-based k-nearest neighbors outlier detection method in large-scale traffic data. In: 2015 IEEE International Conference on Digital Signal Processing (DSP). IEEE, pp. 507–510.

Feng, L.J., Niu, Y.X., Liu, Z.F., Wang, J.F., Zhang, K., 2019. Discovering technology opportunity by keyword-based patent analysis: a hybrid approach of morphology analysis and USIT. Sustainability 12 (1), 136.

Gu, C.J., Huang, L.C., 2009. In: A Systematic Study of the Work that Web Mining does in the Technology Opportunities Analysis. 2009 First International Workshop on Database Technology and Applications. IEEE, pp. 299–302.

Guo, H., Jia, X., Liu, L., Cao, X., Wang, N., Wang, Z.L., 2018. Freestanding triboelectric nanogenerator enables noncontact motion-tracking and positioning. ACS Nano 12 (4), 3461–3467.

Han, X.T., Zhu, D.X., Wang, X.F., Li, J., Qiao, Y.L., 2019. Technology opportunity analysis: combining sao networks and link prediction. IEEE Trans. Eng. Manag. 68 (5), 1288–1298.

Huang, L.C., Wang, J.J., Li, X., Wang, L.Z., 2016. Detecting technology opportunities based on papers and patents for perovskite solar cells. J. Inf. 035 (007), 686–695.

Jang, S., Oh, J.H., 2018. Rapid fabrication of microporous BaTio3/PDMS nanocomposites for triboelectric nanogenerators through one-step microwave irradiation. Sci. Rep. 8 (1), 14287.

Jia, Y.B., Cheng, Y., Du, X.K., Feng, L.J., 2017. Analysis of technology opportunities about the CBM mining method based on multidimensional technology innovation map. Procedia Eng. 174, 251–259.

Kenaza, T., Bennaceur, K., Labed, A., 2018. An efficient hybrid SVDD /clustering approach for anomaly-based intrusion detection. In: Proceedings of the 33rd Annual ACM Symposium on Applied Computing, pp. 435–443.

Kerr, C.I., Mortara, L., Phaal, R., Probert, D.R., 2006. A conceptual model for technology intelligence. Int. J. Technol. Intell. Plan. 2 (1), 73–93.

Khan, L., Awad, M., Thuraisingham, B., 2007. A new intrusion detection system using support vector machines and hierarchical clustering. VLDB J. 16 (4), 507–521.

Khandelwal, G., Minocha, T., Yadav, S.K., Chandrasekhar, A., Raj, N.P.M.J., Gupta, S.C., Kim, S.J., 2019. All edible materials derived biocompatible and biodegradable triboelectric nanogenerator. Nano Energy 65, 104016.

Kim, B., Gazzola, G., Yang, J., Lee, J.M., Coh, B.Y., Jeong, M.K., Jeong, Y.S., 2017. Two-phase edge outlier detection method for technology opportunity discovery. Scientometrics 113 (1), 1–16.

Kim, D., Jin, I.K., Choi, Y.K., 2018. Ferromagnetic nanoparticle-embedded hybrid nanogenerator for harvesting omnidirectional vibration energy. Nanoscale 10 (26), 12276–12283.

Lee, C., Lee, G., 2019. Technology opportunity analysis based on recombinant search: patent landscape analysis for idea generation. Scientometrics 121 (2), 603–632.

Lee, Y., Kim, S.Y., Song, I., Park, Y., Shin, J., 2014. Technology opportunity identification customized to the technological capability of SMEs through two-stage patent analysis. Scientometrics 100 (1), 227–244.

Lee, C., Kang, B., Shin, J., 2015. Novelty-focused patent mapping for technology opportunity analysis. Technol. Forecast. Soc. Chang. 90, 355–365.

Lee, C., Jeon, D., Ahn, J.M., Kwon, O., 2020. Navigating a product landscape for technology opportunity analysis: a word2vec approach using an integrated patent-product database. Technovation 96, 102140.

Li, X., Tao, J., Guo, W., Zhang, X., Luo, J., Chen, M., Pan, C., 2015. A self-powered system based on triboelectric nanogenerators and supercapacitors for metal corrosion prevention. J. Mater. Chem. A 3 (45), 22663–22668.

Li, X., Xie, Q.Q., Huang, L.C., Wu, F.F., 2018. Research on evolution trace of emerging technology based on subject-action-object semantic mining. Science of science and management of. Sci. Technol. 39 (01), 17–31.

Li, X., Xie, Q., Daim, T., Huang, L., 2019. Forecasting technology trends using text mining of the gaps between science and technology: the case of perovskite solar cell technology. Technol. Forecast. Soc. Chang. 146, 432–449.

Li, X., Fan, M.J., Zhou, Y., Fu, J., Huang, L.C., 2020. Monitoring and forecasting the development trends of nanogenerator technology using citation analysis and text mining. Nano Energy 71, 104636.
Li, X., Wang, C., Zhang, X., Sun, W., 2020. Generic SAO similarity measure via extended Sørensen-dice index. IEEE Access 8, 66538–66552.
Liu, F.T., Ting, K.M., Zhou, Z.H., 2012. Isolation-based anomaly detection. ACM Trans. Knowl. Discov. Data 6 (1), 1–39.
Macqueen, J., 1967. Some methods for classification and analysis of multivariate observations. In: Proceedings of the fifth Berkeley symposium on mathematical statistics and probability, 1(14), pp. 281–297.
Meng, X., Cheng, Q., Jiang, X., Fang, Z., Chen, X., Li, S., Li, C., Sun, C., Wang, W., Wang, Z.L., 2018. Triboelectric nanogenerator as a highly sensitive self-powered sensor for driver behavior monitoring. Nano Energy 51, 721–727.
Meyer, M., 2000. Does science push technology?Patents citing scientific literature. Research Policy 29 (3), 409–434.
Moehrle, M., Walter, L., Geritz, A., Müller, S., 2005. Patent-based inventor profiles as a basis for human resource decisions in research and development. R&D Manag. 35 (5), 513–524.
Ogawa, T., Kajikawa, Y., 2015. Assessing the industrial opportunity of academic research with patent relatedness: a case study on polymer electrolyte fuel cells. Technol. Forecast. Soc. Chang. 90, 469–475.
Ou, Y.H., Qu, X.C., Shi, B.J., 2017. Recent progress on nanogenerator for implantable self-powered biomedical systems. In: Life Science Instruments, 15, pp. 3–14.
Pang, Y.K., Chen, S., Chu, Y.H., Wang, Z.L., Cao, C.Y., 2019. Matryoshka-inspired hierarchically structured triboelectric nanogenerators for wave energy harvesting. Nano Energy 66, 104131.
Park, Y., Yoon, J., 2017. Application technology opportunity discovery from technology portfolios: use of patent classification and collaborative filtering. Technol. Forecast. Soc. Chang. 118, 170–183.
Park, I., Yoon, B., 2018. Technology opportunity discovery for technological convergence based on the prediction of technology knowledge flow in a citation network. J. Informet. 12 (4), 1199–1222.
Porter, A.L., Detampel, M.J., 1995. Technology opportunities analysis. Technol. Forecast. Soc. Chang. 49 (3), 237–255.
Rifkin, Jeremy, 2011. The third industrial revolution: how lateral power is transforming energy, the economy, and the world. Macmillan.
Rip, A., 1992. Science and technology as dancing partners. In: Kroes, P., Bakker, M. (Eds.), Technological Development and Science in the Industrial Age. Kluwer Academic, Dordrecht, pp. 231–270.
Roh, T., Jeong, Y., Jang, H., Yoon, B., 2019. Technology opportunity discovery by structuring user needs based on natural language processing and machine learning. PloS ONE 14 (10), e0223404.
Seo, W., Yoon, J., Park, H., Coh, B.Y., Lee, J.M., Kwon, O.J., 2016. Product opportunity identification based on internal capabilities using text mining and association rule mining. Technol. Forecast. Soc. Chang. 105, 94–104.
Shi, X., Cai, L., Song, H., 2019. Discovering potential technology opportunities for fuel cell vehicle firms: a multi-level patent portfolio-based approach. Sustainability 11 (22), 6381.
Shi, L., Dong, S., Ding, P., Chen, J., Liu, S., Huang, S., Huang, X., Farooq, U., Zhang, S., Li, S., Luo, J., 2019. Carbon electrodes enable flat surface PDMS and PA6 triboelectric nanogenerators to achieve significantly enhanced triboelectric performance. Nano Energy 55, 548–557.
Shibata, N., Kajikawa, Y., Sakata, I., 2010. Extracting the commercialization gap between science and technology—case study of a solar cell. Technol. Forecast. Soc. Chang. 77 (7), 1147–1155.
Shibata, N., Kajikawa, Y., Sakata, I., 2010. Opportunity discovery by assessing the gap between science and technology case study of secondary batteries. In: 2010 IEEE International Conference on Management of Innovation & Technology. IEEE, pp. 788–793.
Shibata, N., Kajikawa, Y., Sakata, I., 2011. Detecting potential technological fronts by comparing scientific papers and patents. Foresight 13 (5), 51–60.
Sleator, D.D., Temperley, D., 1995. Parsing english with a link grammar. Department of Computer Science, Carnegie Mellon University, Carnegie Mellon arXiv preprint cmp-lg/9508004.
Sun, J.G., Tan, R., Guo, J., Chai, T., Liu, X., 2015. Analysis on technology opportunities of evolution bifurcations. Procedia Eng. 131, 1113–1120.
Turney, J., 1991. What drives the engines of innovation? New Sci. 132, 35–40.
Wang, S., Lin, L., Wang, Z.L., 2012. Nanoscale triboelectric-effect-enabled energy conversion for sustainably powering portable electronics. Nano Lett. 12 (12), 6339–6346.
Wang, M.Y., Fang, S.C., Chang, Y.H., 2015. Exploring technology opportunities by mining the gaps between science and technology: microalgal biofuels. Technol. Forecast. Soc. Chang. 92, 182–195.
Wang, X., Ren, H., Chen, Y., Liu, Y., Qiao, Y., Huang, Y., 2019. Measuring patent similarity with SAO semantic analysis. Scientometrics 121 (1), 1–23.
Williams, R.A., Edge, W.D., 1996. The social shaping of technology. Res. Policy 25 (6), 865–899.
Xi, Y., Wang, J., Zi, Y., Li, X., Han, C., Cao, X., Hu, C., Wang, Z.L., 2017. High efficient harvesting of underwater ultrasonic wave energy by triboelectric nanogenerator. Nano Energy 38, 101–108.
Xia, K.Q., Zhu, Z.Y., Fu, J.M., Li, Y., Chi, Y., Zhang, H., Du, C.L., Xu, Z., 2019. A triboelectric nanogenerator based on waste tea leaves and packaging bags for powering electronic office supplies and behavior monitoring. Nano Energy 60, 61–71.
Yang, C., Huang, C., Su, J., 2018. An improving SAO network-based method for technology trend analysis: a case study of graphene. J. Informet. 12 (1), 271–286.
Yoon, B., 2008. On the development of a technology intelligence tool for identifying technology opportunity. Expert Syst. Appl. 35 (1–2), 124–135.
Yoon, J., Kim, K., 2012. Detecting signals of new technological opportunities using semantic patent analysis and outlier detection. Scientometrics 90 (2), 445–461.
Yoon, B., Magee, C.L., 2018. Exploring technology opportunities by visualizing patent information based on generative topographic mapping and link prediction. Technol. Forecast. Soc. Chang. 132, 105–117.
Zhang, J.Z., Wen, Q.Y., 2020. Early detection of technology opportunity based on analogy design and phrase semantic representation. Scientometrics 125 (1), 551–576.

Xin Li is a professor in the College of Economics and Management, Beijing University of Technology, China. His recent research interests include technology forecasting, data mining, technology roadmapping, and innovation management. He has published many papers on international journals, such as, Technological Forecasting & Social Change, International Journal of Technology Management, IEEE Transactions on Engineering Management.

Yundi Wu is a master candidate in the College of Economics and Management, Beijing University of Technology, China. Her recent research interests include emerging technologies management, and data mining. She is working on how to identify technology opportunities using data mining.

Haolun Cheng is a master candidate in the College of New Media , Beijing Institute of Graphic Communication, China. His recent research interests include digital twins, data mining and information visualization. He is working on how to build mapping knowledge domain by using data mining and information visualization.

Qianqian Xie is a PhD student in Centre of Science and Technology Studies (CWTS), Leiden university. Her recent research interests focus on bibliometric, text mining, medical health, and emerging technology forecasting.

Tugrul Daim is a professor in the Department of Engineering and Technology Management, Portland State University Portland, OR, USA. His recent research interests include technology forecasting, and technology roadmapping.